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Recent advances in the aerothermodynamics of spiked hypersonic vehicles

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ABSTRACT

Among a variety of design requirements, reducing the drag and aeroheating on hypersonic vehicles is the most crucial one. Unfortunately, these two objectives are often conflicting. On one hand, sharp slender forebodies design reduces the drag and ensures longer ranges and more economic flights. However, they are more vulnerable to aerodynamic heating. On the other hand, blunt forebodies produce more drag, however, they are preferred as far as aeroheating is concerned. In addition, in the context of hypersonic vehicles, blunt geometries are preferred over slender ones for practical implications such as higher volumetric efficiency, better accommodation of crew or on-board equipment.

In principle, a blunt vehicle flying at hypersonic speeds generates a strong bow shock wave ahead of its nose, which is responsible for the high drag and aeroheating levels. There have been a number of efforts devoted towards reducing both the drag and the aeroheating by modifying the flowfield ahead of the vehicle's nose. Of these techniques, using spikes is the simplest and the most reliable technique. A spike is simply a slender rod attached to the stagnation point of the vehicle's nose. The spike replaces the strong bow shock with a system of weaker shocks along with creating a zone of recirculating flow ahead of the forebody thus reducing both drag and aeroheating.

Since their introduction to the high-speed vehicles domain in the late 1940s, spikes have been extensively studied using both experimental facilities and numerical simulation techniques. The present paper is devoted to surveying these studies and illustrating the contributions of the authors in this field. The paper also raises some of the areas in the field that need further investigations.

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1. Introduction

Among a number of design requirements, the reduction of both drag and aerodynamic heating is the major challenge in the design of supersonic and hypersonic vehicles. Reducing the aerodynamic drag on these vehicles ensures reaching the desired range (or altitude) and, in the same time, enables economizing the fuel usage, simplifying the propulsion system requirements, and maximizing the ratio of payload to take-off gross weight [1]. Pointed bodies generate less drag compared with blunt bodies at supersonic and hypersonic conditions. Hence, they provide a good choice as far as drag reduction is concerned. Unfortunately, in the context of high speed vehicles' design, pointed bodies cannot be employed due to the extreme levels of aerodynamic heating associated with these bodies in high speed regimes. High levels of aerodynamic heating can cause malfunction [2] or even damage of the delicate on-board electronic devices [3,4]. In addition, excessive heating can cause ablation to the vehicle material, which yields fluctuations in the vehicle performance [5]. In the presence of high pressure loads, severe heating can cause a complete material failure [6] or at least surface erosion [7]. Employing the conventional thermal protection systems (TPS) complicates the design, increases the burn-out weight, or deteriorates the original design, as in the case of ablating shields. Extreme aeroheating levels experienced by re-entry vehicles further complicate the TPS requirements and result in communication black-out due to ionization of air, a problem that cannot be resolved using TPS. Blunt bodies were found to yield lower heating levels compared with their pointed counterparts since they tend to distribute the aeroheating over a relatively larger area especially at the stagnation point [1,8–10]. Consequently, blunting the forebody is viewed as the primary alternative in hypersonic regimes [11]. In fact, supersonic and hypersonic vehicles such as missiles, interplanetary space missions, space planes, and launch missiles usually have blunt shapes [10,12]. Besides, in some applications, blunting the nose becomes a favored design requirement on its own. For example, blunt noses are recommended for better accommodation and operation of crew or seeker devices (optical, infrared, and radar) with a more efficient usage of the available volume [2,13]. Blunting the forebody of long missiles is sometimes adopted to make the use of shorter launchers and launch tubes more cost-effective [14]. Moreover, the volumetric efficiency of the blunt noses is higher than that of their pointed counterparts [15].

Obviously, designing a vehicle that simultaneously satisfies both minimum drag and aerodynamic heating is not a simple task and there must be a tough trade-off between these two vital requirements. The design of the re-entry vehicles such as reusable launch vehicles and long-range and ballistic missiles is rather complicated. On one hand, it is desired to use a pointed forebody to minimize the drag for less fuel consumption during the take-off (ascent) phase. On the other hand, due to excessive heating loads during the descent phase, a blunt forebody is advantageous to yield the desired vehicle deceleration as well [1,9,10,16].

In the high speed regimes, a blunt forebody yields a strong detached bow shock wave ahead of it. This shock wave is responsible for the elevated pressure and aeroheating levels attained by the downstream flow. Thus, it was believed that the excessive drag and heating loads can be reduced by altering the flowfield pattern ahead of the blunt forebody so as to eliminate the strong shock wave or replace it with a weaker system of shock waves. A variety of techniques have been implemented in this regard. These techniques include spikes and aerodisks, focused gas jet, laser or microwave beams upstream of the nose stagnation point, energy deposition using plasma torch, arc discharge, dc corona discharge, and even supersonic projectiles fired ahead of the blunt forebody [10]. Out of these varieties of techniques, the use of spikes proved to be the simplest and the most effective technique in reducing both drag and aerodynamic heating [15]. It is considered as a compromise of two requirements namely, lower aerodynamic heating for re-entry and lower drag for atmospheric flights [17].

The spike is simply a thin cylindrical rod of fixed length mounted at the stagnation point of the blunt forebody and projected in the upstream direction. In some applications, protruding and telescopic spikes of variable length were implemented [14,18], while in other applications, a spike attached to the re-entry vehicle was ejected in the re-entry phase only [19]. Conceptually, a spike introduces two major modifications to the flowfield upstream of the blunt forebody. Firstly, it replaces the strong detached shock wave with a system of weaker oblique shock waves. Secondly, it acts as a "flow separator"; the spike encourages the separation of the boundary layer from its surface and the creation of a shear layer. The latter propagates downstream, reattaches on the blunt forebody surface, and envelopes a zone of recirculation in which the flow attains low pressure and velocity values. This zone screens a considerable portion of the blunt forebody surface and results in a significant drop in surface pressure and temperature. Only at the zone of shear layer reattachment on the blunt forebody, the local heating rate and the surface temperature attain high values. In addition, to turn the flow outside the shear layer parallel to the forebody surface, a shock wave is created at the reattachment zone. Immediately downstream of this reattachment shock, the flow pressure attains high local values. Nevertheless, the overall effect is a significant reduction in both the drag and the aerodynamic heating.

To further enhance the effectiveness of the spike, a relatively larger tip of a variety of designs, called the aerodisk, can be used. An aerodisk mounted at the tip of a spike of a fixed length has the role of adding further reduction in both drag [20] and aerodynamic heating [21] over a wider range of Mach numbers [6] and incidence angles [16]. It can also compensate the drag reduction in cases when a shorter spike is to be used [15]. The typical features of the flowfield associated with unspiked and aerodisk-spiked blunt bodies in high flow speeds are illustrated in Fig. 1. These features were given various terminologies in the literature. For instance, the oblique shock ahead of the spike/aerodisk tip was referred to as conical shock wave [18,22], bow shock [6], and flow separation shock [10]. Nevertheless, this shock can be conical or bowed (depending on the spike tip/aerodisk geometry) and can take place at or upstream of separation point. Hence, to avoid confusion, the term "foreshock" is introduced to refer to the shock wave at the spike/aerodisk tip whereas the term "separation shock" is used to refer to the shock wave created due to shear layer separation. Similarly, the reattachment shock was referred to as the recompression shock [21]. For the sake of unifying the definitions, the terminologies illustrated in Fig. 1 are suggested.

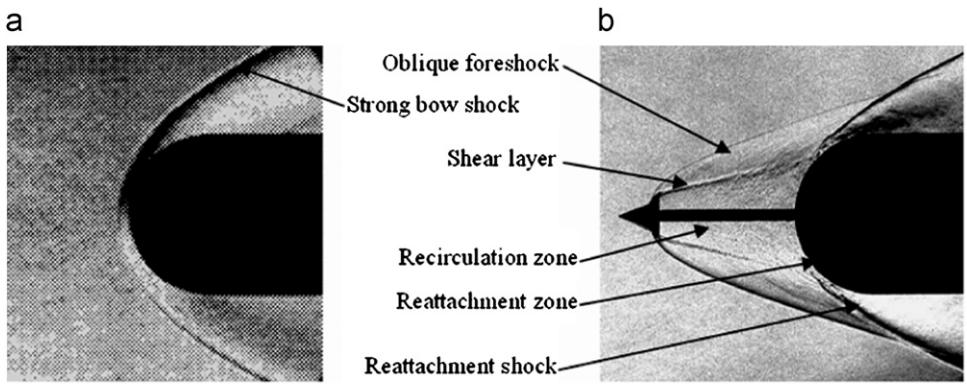

Fig. 1. Macroscopic features of the flowfield ahead of a blunt forebody in hypersonic speeds. (a) without [23] and (b) with the spike/aerodisk [4].

The flow outside the recirculation zone moves past the slender (nearly conical) shear layer and the shoulder of the blunt forebody. Eventually, mounting the spike/aerodisk has the role of virtually transforming the blunt forebody into a more slender "effective" body [19]. As a consequence, the overall fineness ratio of the vehicle increases with a slight weight penalty [13,19]. In addition, mounting the spike leads to a reduction in pressure and heating levels on the rear and base parts of the body as well [11]. It is also known that the use of spikes protects the forebody from other problems associated with elevated temperatures in the hypersonic regime such as dissociation, ionization, and recombination of air. Only at the very vicinity of the spike tip, these reactions have an effect whereas the flowfield around the forebody remains unaffected such that the general laws of perfect gases apply [11], which simplifies the numerical simulation as well. Moreover, at angles of attack, the spike has the effect of increasing the lift [2,19,24] and slightly shifting the vehicle's center of pressure upstream, which enhances its maneuverability [4,25].

The benefits of using spikes with high speed vehicles are not limited to the aerodynamics and flight mechanics features as mentioned above. It is interesting to note that, in the context of high speeds vehicles' design, spikes were also implemented to accomplish other operational requirements. For instance, a spike can be used as a Pitot tube to measure the flight speed. In planetary vehicles, spikes were used as probes to collect gas samples [26] whereas in antitank projectiles, spikes provide a stand-off distance for a more efficient warhead detonation [27]. However, the use of spikes is subject to some limitations. Clearly, the spike produces no significant effect if it is very short [26], in low Mach number values [2,28], or in rarefied conditions [29]. In addition, the spike becomes effective only if the forebody is blunt enough to generate a bow shock wave ahead of it [30]. Moreover, structural and rigidity considerations should be taken into account in cases of very long spikes and measures should be taken to reduce the intensive heating loads at the spike tip [31]. Whether the attachment of the spike at the tip of the blunt forebody nose will affect the function of any seeker devices should also be considered [32]. In addition, the benefits of using the spikes diminish as the incidence angle increases [6,10]. It was thus recommended that the use of spikes should be limited to vehicles that perform low maneuvers during flight [2].

In addition, the use of spikes has a number of potential problems that should be carefully considered. Obviously, the upstream shift of the vehicle's center of pressure reduces its stability [16]. Besides, the pitching moment acting on the vehicle increases, which invokes some additional considerations to compensate [16,21]. Furthermore, the flow separation induced by the spikes was found to cause aerodynamic undamping of some oscillation modes during flight [14]. Moreover, under certain flight conditions and blunt forebody/spike configurations, the flow can become highly unstable, which invokes some problems such as fluctuations in surface pressure and heat flux [19], elevated acoustic and structural dynamic loads [18], bending of the spike [14], and flight disturbances and control complications [24,25]. These problems can get rather worse due to the hysteresis of flow separation taking place during the extension and retraction of protruding and telescopic spikes [18,33]. In addition, under certain conditions, parts of the blunt forebody downstream of the reattachment zone are exposed to excessive aerodynamic heating loads [19]. In fact, aerodynamic heating reduction is not always guaranteed and, in some cases, the total heat transfer to the blunt forebody can be doubled in the presence of the spike mainly due to the elevated heating in the reattachment zone [34].

The purpose of the present review paper is to survey the efforts that have be made in the field of spiked bodies aerothermodynamics, which will be discussed in the next section. The paper then introduces the contributions of the authors in this field. The paper finalizes with some conclusions and recommendations for future studies.

2. Literature review

In the open literature, a huge body of studies concerning the spiked bodies aerothermodynamics can be found. Almost all these studies are discussed in the present section and summarized in Appendix. For a clearer survey, these studies are classified into four categories. The studies focusing on the general features of the spiked bodies aerothermodynamics are illustrated first, the experimental studies followed by the numerical simulation studies. Next, the studies (both experimental and numerical) concerning the flow unsteadiness associated with the spiked bodies are discussed. Finally, some interesting studies of the non-conventional usages of spikes are presented. In what follows, unless otherwise stated, the value of Reynolds number is calculated based on the freestream conditions and the main body's diameter.

2.1. Experimental studies

The studies related to the spiked bodies' aerodynamics can be traced back to 1947 [13]. In the Langley Pilotless Aircraft Research Division, a number of firing tests were executed on blunt and sharp nosed bodies in Mach numbers 0.95–1.37. It was proposed that mounting a thin rod with a relatively larger conical tip (the aerodisk, called the "windshield" at that time) on the blunt forebody could increase the effective fineness ratio of the latter

with a slight weight penalty. Based on firing tests measurements, Alexander [13] reported a slight drag reduction using the conical "windshield", which was still higher than that of the pointed forebody in Mach 1.37. A step forward was made by Moeckel [35] four years later. He measured the surface pressure on a spiked parabolic forebody with a hemispherical nose in a Reynolds $2.\times10^5$ (based on the nose diameter), Mach 1.76, 1.93 freestream. A plain conically pointed spike (called the "projecting rod") whose length varied from 1 to 12 times the forebody nose diameter was used.

The comprehensive study by Mair in 1952 [36] is probably the first landmark study in the field. He experimentally examined the flow around spiked flat cylindrical and hemisphere cylindrical models in a Mach 1.96, Reynolds 1.65×10^5 flow. A conically pointed spike (called the "probe") was utilized and its length-to-model diameter ratio, L/D, was varied up to 6. Mair presented a very fine anatomy of the flowfield around the spiked bodies and thoroughly discussed the impact of the spike length on the flowfield. He also measured the surface pressure and the corresponding drag on a model with $L/D=1.44$. Mair was the first to record a sign of flow instability around spiked bodies. For the flat cylindrical models and for a certain range of spike length, a significant flow instability is invoked in which the shock wave at the spike tip changes its shape in a self-sustained manner. Mair proposed the first explanation of this form of "flow oscillation" based on the pressure difference between the flow downstream of the reattachment shock and the flow inside the recirculation zone. In the same year, Jones [37] investigated the flow around conical models with a hemispherical nose at a Mach of 2.72. A sharp pointed "protruding rod", which length varied from 1 to 6 times the forebody nose radius, was used. Jones was also the first to point out that only one streamline can stagnate on the forebody surface at the reattachment point, this streamline was not given a specific term.

The term "spike" was first used in 1954 by Piland and Putland [28]. They analyzed the results of firing tests in Mach numbers ranging from 0.7 to 1.3 and Reynolds numbers from $1.44{:}2.64\times10^5$ flight conditions. The study involved (among others) projectiles of hemispherical noses with and without sharp pointed spikes of two different lengths. They concluded that no drag reduction was achieved for that range of Mach numbers regardless of the spike length. Stalder and Nielsen [34] conducted the first investigation of the aerothermodynamic effects of the spikes. They measured the heat transfer to a hemisphere cylindrical model in a Mach number ranging from 0.12 to 5.04 and the Reynolds number ranging from $1.55{:}9.85\times10^5$. The model was equipped with spikes of pointed tip, flat aerodisk, and conical aerodisk and their L/D ratio varied from 0.5 to 2. They reported that, regardless of the spike length and tip geometry, heat transfer to the spiked model was as twice as that of the unspiked model. They argued that the cause of heat transfer rise was the high turbulence level in the dead air zone and the impingement of turbulent shear layer on the model surface.

Jones [38] used the "windshield" to reduce the drag on a hemisphere conical model in a Mach 2.72, Reynolds 1.83×10^6 flow. The "rod" was 0.415, 1.125, and 1.464 times the model nose diameter. To study the role of their shapes, conical windshields with variable lengths and angles were examined. The drag reduction was found to depend on both rod length and windshield geometry and for long rods, the windshield was found to lose its benefits in reducing the drag. Beastall and Turner [33] conducted an investigation on flat cylindrical models in a Reynolds 1.125×10^6, Mach 1.5, 1.6, and 1.8 flow. The models were equipped with spikes of pointed ogive tips along with conical aerodisks of variable vertex angles. A protruding spike with L/D up to 4.7 was used. They reported a hysteresis phenomenon in the flowfield evolution. The drag on the models for a specific spike length was found to depend on whether the spike was extended or retracted. Apart from the "violent oscillations" first recorder by Mair, Beastall and Turner recorded a new type of flow unsteadiness associated with longer spikes. This "mild oscillation" type was characterized by small oscillations of the shear layer. Hermach et al. [39] used the term "pulsating flow" to refer to the violent oscillations. They focused on the impact of this type of flow unsteadiness on the aerothermal characteristics of the spiked bodies. Pointed spikes of variable tip design and L/D ratio up to 2 were fitted on a flat cylindrical model in a Mach 3.5, Reynolds 1.73×10^6 flow. It was found that, at certain optimum spike length, pulsating flow could produce up to 22% reduction in the temperature recovery factor compared to the laminar boundary layer. In addition, the frequency of pulsating flow was found to monotonically decrease by increasing the spike length. Bogdonoff and Vas [11] undertook an important experimental investigation of pressure and heat transfer distribution on spiked bodies. Two models namely, flat cylindrical and hemisphere cylindrical models, were tested in a Mach 14, Reynolds 3.6×10^5 Helium flow. The role of varying the length of a pointed spike (L/D up to 8) was investigated. They argued that the heat transfer to the spiked model is significantly reduced if the separated shear layer remains laminar until reattachment. Both types of flow unsteadiness were recorded for flat cylindrical models.

Another milestone in the field was set by Crawford in 1959 [23]. He conducted an extensive experimental investigation of drag and aerodynamic heating on a spiked hemisphere cylindrical model in Mach 6.8 flow. The model was equipped with a pointed spike of variable length (L/D up to 4). The role of Reynolds number was first investigated in this study by varying its value from 0.12×10^6 to 1.5×10^6 based on freestream conditions and model diameter. In all cases, the model's surface pressure and heat flux attained peak values at the reattachment point. The pressure peak remained well below the stagnation pressure whereas the peak heat flux ranged from half (at the lower bound of Reynolds number) to double (at the upper bound) the heat transfer value at the stagnation point of the unspiked model for all spike lengths. Maull [9] conducted the first attempt to address the impact of modifying the body shape on the flowfield structure. He investigated the role of rounding the shoulder of a spiked flat cylindrical model on the flow oscillations in Mach 6.8, Reynolds 8.5×10^4 flow. A pointed spike of a variable length (L/D up to 3) was used and the roundness (ratio of shoulder radius to model diameter) of the models' shoulders varied from 0 (sharp shoulder) to 0.5 (hemisphere cylindrical model). Maull refined the mechanism of boundary layer separation over the spike length based on the pressure equilibrium on both sides of the separation shock wave. Maull also clarified the mechanism of flow stability based on the mass flow equilibrium between the flow scavenged by the shear layer and that reversed inside the dead air zone. The governing factor was argued to be the flow turning angle on the model's face at reattachment. A hemisphere cylindrical model was found to yield a stable flow regardless to the spike length.

Another landmark was established by Wood [22]. He was the first to investigate the variation in the flowfield structure by changing both the model geometry and the spike length in Mach 10 and Reynolds number 0.5×10^5 freestream. The model was a cone cylindrical model, which vertex angle varied from 30° to 180° (flat cylinder), while the used pointed spike had a variable length (L/D up to 5). Wood classified the flow pattern around spiked conical models into five categories depending on the spike length and the model's semi-apex angle. Wood was the first to use the term "dividing streamline", first introduced by Chapman et al. [40], to refer to the streamline that links the separation and reattachment points of the share layer at steady flow conditions.

The role of spikes in reducing drag and heat transfer was also supported by Wagner and Pine [8] in a Mach 19.4, Reynolds 0.3×10^6 with Helium flow. Among other models representing the common re-entry vehicle designs, blunt parabolic cylindrical and hemisphere cylindrical models were equipped with a pointed spike of variable length (L/D up to 5). The performance of spikes at incidence in very low Reynolds number environments (similar to those experienced by planetary vehicles) was investigated by Sims et al. [29]. The drag on two models namely, flat cylinder and highly blunt cone in a Mach 10.1, Reynolds 194 Nitrogen flow, was measured. The models were equipped with a pointed spike of variable length (L/D up to 5).

Another important landmark in the field was established by Holden [30]. In this fine experimental investigation, focus was made on the heat transfer to spiked models in Reynolds 0.27×10^6, Mach 10 and 15 flow. Three models namely, hemisphere cylinder, flat cylinder, and cone cylinder, were investigated. The semi-vertex angle of the conical nose had the values of 30°, 45°, 60°, and 75° while the L/D ratio of the pointed spike varied up to 4. For cone cylindrical models, Holden extended and refined the classification of flow patterns around spiked models first established by Wood [22] to higher Reynolds and Mach number values. He also recorded the mild oscillations on conical models for the first time and added a new pattern corresponding to it. Holden showed that the peak local heat transfer rate at the reattachment point is in a direct proportionality with reattachment angle. The latter is dictated by both the spike length and the cone vertex angle.

Khlebnikov [41] focused on measuring the heat transfer to spiked spherical models in a Mach 3, Reynolds 1.4×10^6 flow. The model was equipped with spikes of different lengths ($L/D=0.283$:1.78) whereas conical and pyramidal aerodisks were fitted at their tips. It was concluded that, the value of peak heat flux varies inversely with its distance from the spike root. McGhee and Staylor [26] investigated the role of mounting very short pointed spikes to blunt conical models in Reynolds 1.83×10^6, Mach 3, 4, and 6 flow. Their experiments involved two conical models with a vertex angle of 120°, blunt nose radii of 0.25, and 0.5 the base radius. The spike length varied from 0.055 to 0.083 depending on the model's base diameter. Three spikes with different conical tips (20°, 30°, and 40° semi-vertex angles) and two different diameters (0.044 and 0.011 of the model base diameter) were tested and the role of 0°, 2°, and 5° incidence was investigated as well. With such short spikes, the flow features were correlated to the ratio of spike length to stand-off distance of the bow shock wave ahead of the unspiked model. The impacts of both spike tip angle and diameter on pressure distribution on the models were found insignificant. Staylor [42] extended this investigation to Mach 9.6, Reynolds 6×10^4:23×10^4 flow and the study included results of up to Mach 20 flow. A third model of a pointed nose was added, the spike length was extended up to 4 times the stand-off distance of the unspiked model, and focus was made on the heat transfer characteristics on the models. The role of spike length was recently confirmed by Humieres and Stollery [43]. They experimentally investigated a re-entry vehicle model of a spherical cap and conical afterbody in a Mach 8.2, Reynolds 9×10^6 flow. They quantitatively confirmed the similarity between spiked bodies and solid cones first addressed by Mair [36].

After the studies of Alexander [13] and Jones [38], the superiority of the aerodisks in drag reduction compared with the plain spike was explained and discussed in many studies [16,20,21]. In [20], a blunt ogive cylindrical model was equipped with an extensible "telescoping" spike with a blunt aerodisk at its tip. The freestream conditions corresponded to transonic (Mach up to 1.2, Reynolds 8:12×10^6) and supersonic (Mach up to 3, Reynolds 4.7:7×10^6) flow. It was proposed that, the further drag reduction is caused by the merging of two dead-air zones: the one ahead of the main body and the one generated at the wake of the aerodisk.

In the rich experimental investigation by Motoyama et al. [21], pressure and heat flux distributions along a spiked hemisphere cylindrical model in a Mach 7, Reynolds 4×10^5 flow were recorded. Moreover, the variation of drag and pitching moment coefficients with incidence angles (up to 8°) was measured. In addition to a plain spike with conical, rounded, and flat ends, hemispherical and flat aerodisks of two different sizes and two spike length ($L/D=0.5$ and 1.0) were tested. Contributions of both the model and the spike/aerodisk in the total pressure drag were also separately addressed. The study focused on the impact of the spike length and tip design on the heat transfer to the main body. The comprehensive investigation by Kalimuthu et al. [16] also confirmed the advantages of using aerodisks. Drag, lift, and pitching moment acting on a hemisphere cylindrical model in a Mach 6, Reynolds 9.44×10^5 with incidence (up to 8°) freestream were measured. The spikes used had a variable length ($L/D=1.5$ and 2) and, in addition to a pointed plain tip, hemispherical and flat aerodisks were examined as well.

Even without using an aerodisk, modifying the tip of a plain spike was proved to have a strong impact on its performance. For instance, Milicev and Pavlovic [25] and Milicev et al. [44] experimentally examined a hemisphere cylindrical model in a Mach 1.89, Reynolds 3.8×10^5 flow at incidence (from −4° to 10°). The model was equipped with four spikes having the same length ($L/D=1$) and different shapes namely, pointed cylindrical, rounded cylindrical, and conical (with two different vertex angles). It was concluded that the spike with a rounded tip produces that maximum drag reduction. Recently, Kalimuthu et al. [45] undertook a similar study. A hemisphere cylindrical model was studied in a Mach 6. The model was equipped with a spike of a rounded tip and ($L/D=1.5$ and 2.0) and the variation of lift, drag, and moment coefficient with incidence angle (up to 8°) was investigated. Huebner et al. [6] focused on finding the range of incidence angles within which the spike would perform the best in terms of drag and aerodynamic heating reduction. The pressure and wall temperature distribution along a cylindrical model capped with a smaller hemisphere was thoroughly measured. The freestream conditions corresponded to a Mach 6.06, Reynolds 2.67×10^6 flow and the incidence angle varied from 0° to 40°. A flat aerodisk was fitted at the tip of the spike that had an L/D ratio of 3.

2.2. Numerical simulation studies

Numerical studies of the spiked bodies started as early as 1980s. Myshenkov [46] applied the solution of the complete Navier–Stokes equations using the finite-difference approach in a two-dimensional axisymmetric computational domain. The freestream conditions corresponded to Mach 0.5: 3 Reynolds 1×10^3:1×10^4 and Mach 2, Reynolds 3.6: 1×10^5 flows. The model was a flat cylinder equipped with a pointed spike with a variable length (up to $L/D=3.25$) and the roles of Mach number, Reynolds number, and spike length were discussed. Paskonov and Cheraneva [47] solved the complete unsteady Navier–Stokes equations in a two-dimensional axisymmetric flowfield around cone cylindrical (semi-vertex angles of 70°) and flat cylindrical models equipped with pointed spikes of variable length (up to $L/D=1$). The freestream conditions corresponded to Mach 2:6, Reynolds 100:500 flows. Karlovskii and Sakharov [48] solved the unsteady Euler equations of an inviscid flow in a two-dimensional axisymmetric computational domain. The model had a truncated conical design whereas the spike had a hemispherical tip and a variable length (up to $L/D=1$). The freestream

conditions corresponded to Mach 3:8 flows. Shoemaker [2] solved the two-dimensional unsteady Navier–Stokes equations and the results were validated with some experimental data on blunt bodies. Shoemaker studied two models namely, blunt cone cylindrical and hemisphere cylindrical in a Mach 2.5 freestream conditions. The spike had a variable length ($L/D=2{:}9$) and a biconical aerodisk at its tip.

Turbulence models were implemented in the numerical scheme by the 1990s. The algebraic Baldwin–Lomax two-layer zero-equation model was widely used at that time [3,27,49,50]. White [3] adopted the finite volume approach to solve the Navier–Stokes equations in the axisymmetric domain around two spiked models in Mach 4 farfield conditions. The models had hemisphere cylindrical and hemisphere capped cylindrical designs and the flat spike used had a variable length ($L/D=0.875$and 1.3) and the results were validated with experimental measurements on the same models. Mikhail [27,49] solved the unsteady Navier–Stokes equations in the two-dimensional axisymmetric flowfield around a complete spiked projectile in a Mach 1.72. The spike had flat and conical tips and its L/D ratios were 1.08 and 1.47, respectively. Results were validated with experimental measurements on the same projectile. Hur et al. [50] solved the complete time-dependent Navier–Stokes equations around complete spiked projectiles with fins. The projectiles had hemisphere cylindrical and flat cylindrical design equipped with flat spike whose L/D ratio was 1.42 and the freestream conditions corresponded to Mach 2.23, Reynolds 1.716×10^6, and Mach 3, Reynolds 7×10^6, zero incidence flow. Density contours and velocity vectors were calculated and the results were validated with experimental data on the same model.

The results of the rich experimental investigation by Crawford [23] were used to validate the numerical results by many researchers. In their comprehensive numerical investigation, Yamauchi et al. [51] tested a spiked hemisphere cylindrical model in Mach 2.01, 4.15, and 6.8, Reynolds 1.4×10^5 flow. The model was equipped with a plain pointed spike with a variable length ($L/D=0.5$, 1, and 2) and the Navier–Stokes equations were solved assuming laminar steady flow conditions on an axisymmetric computational domain. Mehta conducted a number of numerical studies [5,7,52] on the same model namely, a hemisphere cylindrical model equipped with a pointed spike. In all the studies, he solved the unsteady Navier–Stokes equations in an axisymmetric computational domain and the flow was assumed laminar. In [5,7], he reproduced the same numerical investigation by Yamauchi et al. and made validations with their results. Asif et al. [53] reproduced numerically the experimental investigations of Crawford [23] and Milicev and Pavlovic [25]. They also compared their results with those of Mehta [5] and Yamauchi et al. [51]. Gauer and Paull [15] used the results of Crawford [23] to validate their extensive numerical investigation. The model tested had a blunt conical shape and the spike had a variable length (L/D varied from 1 to 4). In addition to pointed and rounded tips, a hemispherical aerodisk (with two different sizes) was attached to the spike tip. The farfield conditions corresponded to Mach 5, 7, and 10 freestream conditions. And the results were also validated with the calculations in [5,51]. In addition to Crawford's experiments, many other experimental investigations were reproduced numerically. For instance, Srivinasan and Chamberlain [54] reproduced numerically the experimental investigation of Reding and Jecmen [55]. Similarly, Kurbatskii and Montanari [24] reproduced the experiments of Gnemmi et al. [4] and Huebner et al. [6]. Mehta [12] reproduced the experiment conducted in [16] using a laminar, steady, axisymmetric numerical model.

The commercial CFD codes were also used to assist and support the experimental results conducted by many researchers such as Gnemmi et al. [4] and Srulijes et al. [57] and the co-researchers of P.K. Reddy. [1,10,56,58]. Gnemmi et al. [4] and Srulijes et al. [57] experimentally studied the flow around spiked hemisphere cylindrical models in a Mach 4.5 flow at incidence (up to 24°). The spike had a fixed length ($L/D=1$) whereas flat, spherical, and biconical aerodisks fitted at its tip were examined. The experiment was reproduced numerically in a three-dimensional computational domain around the models. The co-researchers of Reddy focused on highly blunt spiked models in a Mach 5.75, Reynolds 1.5×10^5 freestream conditions. In [1], the model was equipped with a flat aerodisk and the spike had a fixed length ($L/D=1$). They recorded the variation of drag coefficient with the incidence angle (up to 12°) and the results were validated with numerical simulation results. The study was extended in [10] to include more spike tip designs namely, hemispherical aerodisk, and flat and pointed plain spike. The spike had a variable length ($L/D=0.2$, 0.5, 0.7, and 1). The variation of surface pressure and heat flux with spike length and the impact of incidence angle on the total drag were discussed and the results were reproduced numerically. The study was extended to a Mach 8 freestream on the same model at zero incidence in [58]. The study focused on the drag reduction only and the results were compared with a numerical simulation. In [56], focus was made on the flow unsteadiness around the same model. The accuracy of the commercial CFD codes in predicting the flowfield features around spiked models was discussed in the interesting work by Roveda [59]. His study focused on comparing the performance of Cobalt and CFD++ codes against Fluent simulation results by Kurbatskii and Montanari [24] and the experimental findings by Heubner et al. [6] and Gnemmi et al. [4].

2.3. Studies focusing on the flow unsteadiness around spiked bodies

In the literature, a number of studies were completely devoted to investigating the flow instability around spiked bodies; they are discussed below.

The two distinct modes of flow instability associated with spike blunt bodies namely, violent and mild oscillations, were given new terminologies by Kabelitz [60] in 1971. The violent mode was termed the "pulsation mode" and the mild mode was termed the "oscillation mode". However, in the Russian literature for some years later, oscillation and pulsation modes were termed as pulsation of the first kind and pulsation of the second kind, respectively. For instance, Antonov and his co-researchers conducted a number of experimental investigations on unsteady flow around spiked models [61,62–64]. In [61], they investigated the flow around a spiked flat cylindrical model in a Mach 6, Reynolds 1×10^6 freestream. The model was equipped with a pointed spike of a varying length (L/D up to 3) and the study focused on pressure and heat transfer fluctuations on the model surface in both modes of unsteadiness. The most important finding of this study is the determination of the range of each mode of unsteadiness. The first kind (oscillation) takes place when $3>L/D>1.4$, while the second kind (pulsation) takes place for $0.3\le L/D\le1.4$ and the flow becomes steady elsewhere. This investigation was extended to Mach 2–6, Reynolds 6×10^4–3×10^6 flow in [62,63]. The model tested had a cone cylindrical shape with the cone semi-vertex angle varying from 45° to 120°. The L/D ratio of the pointed spike used varied up to 5 whereas its diameter varied from 0.06 to 0.3 of the model diameter.

A landmark study was conducted by Antonov et al. [64], in which they implemented the same freestream conditions and used the same models of their previous studies. In this work, they managed to capture the evolution of pulsation flow mode for the first time and determined its cycle, mechanism, and cause based on scientific evidence. Kenworthy [65] also examined a flat

cylindrical model in a Mach 2.1, Reynolds 0.12×10^6 flow. The model was equipped with a pointed spike with $L/D=1$. In addition, Panaras [66,67] independently proposed the same conclusions as Antonov et al. and proposed the same description and explanations. The cause of pulsation as explained by Antonov et al. [64] and Kenworthy [65] is different and more comprehensive than that was first proposed by Mair [36] and prevailed in the later studies. In fact, the two explanations are based on the feeding or air from an external source. The only difference lies in the source of this air supply. While Mair assumed that the air is coming directly from behind the bow shock, Antonov et al. proved and Kenworthy confirmed that the air is coming from behind the oblique separation shock. Panaras highlighted that the flow behind the bow shock is definitely subsonic whereas the flow inside the separation zone is supersonic. Hence, the only possible source of the "filling" flow is from behind the weak oblique shock wave. Consistent with [61], Panaras [66] showed that pulsation mode is switched to oscillation mode if the spike length exceeds 1.4 times the model diameter (in a Mach 6 flow). The same explanation of pulsation mode was confirmed by Zapryagaev and Mironov [68] and Zapryagaev and Kavun [69], who investigated flat cylindrical models equipped with pointed spikes of L/D ratio varying up to 1.5. In [68], freestream conditions corresponding to Mach 2 and 3 and Reynolds $2.4{:}5\times10^6$ were applied and wereextended in [69] to a Mach 6.08, Reynolds 6.35×10^5 flow. The pulsation mode was also investigated by Calarese and Hankey [18], Shang et al. [70], and Shang and Hankey [71]. They studied the Mach 3, Reynolds 4×10^5 flow around spiked truncated conical model (cone half angle is 9°) and the spike used had a hemispherical tip and its L/D ratio was 0.75. The most important contribution of this group of researchers is the implementation of numerical techniques (for the first time) in solving the flowfield around spiked models. In [70,71], they solved the unsteady laminar Navier–Stokes equations on the two-dimensional axisymmetric flowfield using the finite difference approach and the results were validated with their own experimental findings. For this model, the pulsation mode was found to occur in the range $0.875\geq L/D>0.25$. The calculated mean pressure at the model shoulder increased with the spike length, reached a maximum (about 17 times the freestream pressure) at $L/D=0.875$, and dropped rapidly afterwards. The density contours and velocity profiles were also calculated during a complete period of pulsation. In [18], an experimental investigation was conducted on the same model with focus on the effect of using an extensible spike. A hysteresis phenomenon was found to take place in the range $0.85\geq L/D>0.7$.

The most important contribution of Kenworthy [65] is the explanation of the cycle, mechanism, and cause of the oscillation mode. According to Kenworthy, the cause of flow oscillation mode is a continuous pressure imbalance on both sides on the shear layer at the reattachment point. Hence, a mass balance is established between the air scavenged out from the separation zone and the air reversed into it. Feszty et al. [72–75] reproduced numerically the experimental investigation of Kenworthy [65]. They implemented the solution of the unsteady laminar Navier–Stokes equations in an axisymmetric domain around the spiked model. In [72], the pulsation mode was investigated. The freestream conditions corresponded to Mach 2.21, Reynolds 0.12×10^6 while the L/D ratio was 1. In [74], the pulsation mode around the same model in a Mach 6, Reynolds 0.13×10^6 flow was investigated. Focus was made on flow oscillation mode in [73]. The same model with a longer spike ($L/D=2$) in a Mach 6, Reynolds 0.13×10^6 flow was investigated. The explanation of the flow oscillation mode and its mechanism as proposed by Kenworthy [65] were confirmed using the numerical tools. The hysteresis phenomena recorded by Kenworthy [65] was reproduced numerically in [75]. The freestream conditions corresponded to Mach 2.21, Reynolds 0.12×10^6 while the L/D ratio of the extensible spike was ranging from 1.25 to 2.4. Panaras and Drikakis [17] reproduced numerically the experiments of Panaras [67], Kenworthy [65], Calarese and Hankey [18], and Crawford [23]. The features of oscillation, pulsation, and hysteresis flow were confirmed.

2.4. Other interesting, non-conventional studies

Apart from the "main stream" studies illustrated above, a number of studies introduced some different implementations of spikes with blunt bodies and possible techniques to further enhance their performance. In what follows, these studies are discussed. Instead of putting the whole model at incidence, Thurman [76] showed that lift could be generated by slightly deflecting the spike alone with respect to the model's symmetry axis. He measured both lift and drag acting on a hemisphere cylindrical model equipped with a pointed spike ($L/D=4$) in a Mach 11.76, Reynolds 1.4×10^4 flow and the spike deflection varied up to 7°. A more advanced study based on almost the same idea was conducted by Schulein [77]. He introduced the concept of "pivoting spike" in which the spike is maintained aligned with the freestream direction while the whole body is at incidence. He experimentally examined the pivoting spike in Mach 2, 3, and 5 with up to 30° incidence. Instead of using a spike, Guy et al. [32] suggested attaching the aerodisk at a distance from the model using a strut thus clearing the nose area of the model; such design was recommended in some practical applications (such as missiles with heat seeker devices). The study involved a hemisphere cylindrical model and a number of conical aerodisks of different vertex-angles and base diameters in Mach 0.92 and 1.62 freestream conditions.

To further enhance the effectiveness on spikes, Khlebnikov [78–80] suggested rotating the spike about its axis. Flat cylindrical, conical, and spherical models were examined in Mach 3, Reynolds $0.8{:}1.6\times10^6$ flow conditions and the L/D ratio varied up to 1.68. A plain pointed spike and a wedge-shaped aerodisk, and incidence angles 0° and 10° were tested. Another technique was suggested to further boost the effectiveness of the spikes namely, the addition of external injection/ burning of gas jet. This technique was first introduced by Reding and Jecmen [55]. They experimentally investigated hemisphere cylindrical and blunt ogive cylindrical models in a Mach 2.2, Reynolds 5.23×10^6 freestream. The L/D ratio of the used spike varied from 1 to 1.875 and a flat aerodisk was fitted at the spike tip. Lateral jets of hydrogen were injected from orifices in the side face of the aerodisk. A similar (but briefer) investigation was conducted by Jiang et al. [31]. A hemisphere cylindrical model equipped with a blunt tipped spike ($L/D=1$) was examined in a Mach 6, Reynolds 1.98×10^7freestream at zero and 4° incidence. A lateral jet of air was applied at the shoulder of the spike's rounded tip. It was believed [31] that the use of jet broadens the range of incidence angles within which the spike will be effective in reducing the drag (such as that discussed by Huebner et al. [6]).

3. Recent contributions

This section reviews recent contributions to the topic by the authors. They conducted a research on the spiked hypersonic vehicles using numerical simulation techniques [81]. The commercial CFD code, Fluent [82], was implemented in this research as the flow solver. The objective of the research was to find the optimum design of such vehicles for both minimum drag and aerothermal effects. The ultimate objective of the research was to

find the optimum design of such vehicles for both minimum drag and aeroheating effects. The case study was a forebody of a blunt hypersonic vehicle flying with a Mach 6, zero incidence, and at an altitude of 60 km in a standard atmosphere. In the context of this research, a thorough investigation of the aerodynamic characteristics was done and a number of new findings were revealed. In the present section, these findings are illustrated first. The results of the design optimization are discussed next.

3.1. Investigation of the aerothermodynamic characteristics of spiked blunt bodies

The investigation of the aerothermodynamic characteristics is conducted on a model of a hemispherical shape of a unit diameter, $D=1$, representing the forebody of the hypersonic vehicle. A spike of a variable length and a diameter of $d=0.05D$ is used and is equipped with a hemispherical aerodisk of a variable diameter. The spike length and the aerodisk diameter are varied such that the overall length of the spike–aerodisk assembly ranges from $0.5D$ to $2.5D$ and the aerodisk diameter ranges from 0 to $0.4D$ with a step 0.05. An aerodisk of zero diameter corresponds to a sharp-pointed spike whereas a diameter of $0.05D$ corresponds to a plain spike with a rounded tip. A total of 45 models were investigated. The geometry of the model is shown in Fig. 2.

The numerical simulation is conducted on an axisymmetric computational domain, which is discretized into a structured grid using Gambit [83]. The sensitivity of the numerical solution to the grid resolution is studied. The validity of the numerical simulation scheme to this type of problems is addressed by comparing it with the experimental results of Crawford [23], Gnemmi et al. [4], and Kalimuthu et al. [16]. Both the grid sensitivity checks and the validation tests are successful and show good results [81].

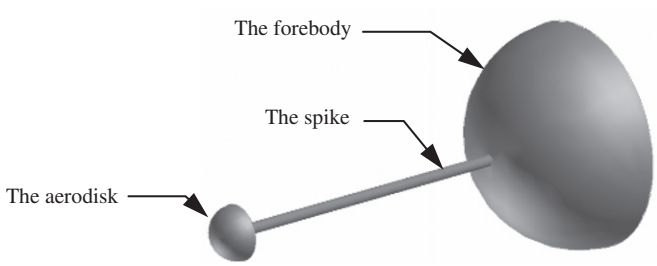

Fig. 2. Geometry of the test model.

3.2. Mechanism of drag reduction, the effective body approach

The drag reduction experienced by spiked blunt bodies is commonly explained by the size of the recirculation zone, which screens a considerable portion of the forebody [10,16,51]. Increasing the spike length causes the reattachment point to move further downstream along the forebody surface, the recirculation zone to expand covering a larger area of the forebody. In addition, the strength of the reattachment shock decreases as the reattachment point moves further towards the forebody shoulder (as a consequence of increasing the spike length). Hence, the surface pressure at the reattachment point is reduced. Overall, the drag is further reduced as the spike length increases. This explanation was widely adopted and was found to sufficiently explain the performance of spiked bodies. However, this explanation fails to address a number of phenomena associated with sharp-pointed spikes. Fig. 3 illustrates the pressure variation along the spike surface and forebody surface for three models with sharp-pointed spikes. The horizontal axis in Fig. 3a represents the normalized distance measured from the conical tip shoulder whereas in Fig. 3b it represents the angle from the model axis measured in the clockwise direction. In both the figures, the local pressure is normalized with respect to the stagnation pressure of the unspiked body at the same freestream conditions, P_{ref}.

Firstly, increasing the spike length reduces the flow pressure *everywhere* and not only at the reattachment point. Secondly, the slight shift of the reattachment point seems insufficient to explain the significant reduction in the peak pressure. The concept of the effective body is implemented by the authors [84] to explain more accurately the mechanism of drag reduction in spiked bodies. In fact, the concept of the effective body is well-established in the field of spiked bodies aerodynamics . It was first addressed by Alexander [13] and simply states that the spiked blunt body is effectively transformed into a more slender body due to the generation of the shear layer. For sharp-pointed bodies, Mair [36] argued that the shape of the recirculation zone was nearly conical and hence the flow outside the shear layer was very similar to a conical flow. He confirmed this explanation by comparing the inclination angle of separation shock with that

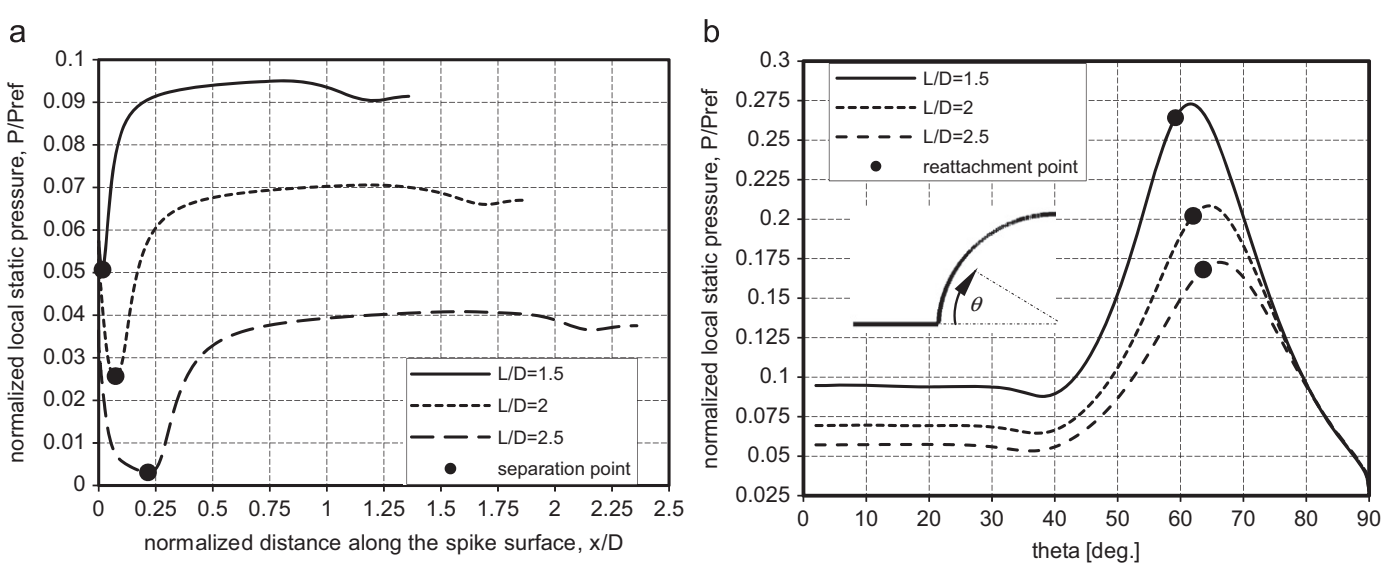

Fig. 3. Surface pressure variation on three different models with pointed spikes: (a) along the spike surface and (b) along the forebody surface.

ahead of a conical body having the same deflection as the shear layer. These angles were found to be in a close similarity. This hypothesis was supported by Crawford [23], Yamauchi et al. [51], and was recently confirmed quantitatively by Humieres and Stollery [43].

However, defining the effective body by the shear layer is not precise. This is because the thickness of the shear layer increases significantly with distance from the separation point [40]. In addition, at reattachment, part of the streamlines constituting the shear layer continues moving past the reattachment zone and merges with the flow downstream of the reattachment shock whereas others are reversed into the recirculation zone. The streamline that separates the "escaping" and "reversing" streamlines was defined by Chapman et al. [40] as the dividing streamline. Wood [22] emphasized that, at equilibrium, the dividing streamline links the separation and reattachment stagnation points. In addition, it was shown [65] that the stability of this streamline dictates the overall stability of the flowfield. Fig. 4 illustrates more details about the dividing streamline of a model with a sharp-pointed spike ($L/D=2$).

The pressure contours in Fig. 4a show that the locations of both separation and reattachment shocks are associated with both ends of the dividing streamline. The dividing streamline is the only streamline in the shear layer that completely stagnates on the body surface at the reattachment point as shown in Fig. 4b. Hence, we find it more accurate to use the dividing streamline rather than the shear layer to define the outer profile of the effective body. To validate this hypothesis, the dividing streamlines for three models with pointed spikes namely, with $L/D=1.5$, 2, and 2.5 are investigated. The inclination angles of dividing streamline and separation shock are compared with the theoretical analysis of supersonic flow around conical bodies [85] in Fig. 5. The comparison shows good agreement of the hypothesis with the theoretical relations.

It is clear from Fig. 5 that increasing the spike length yields a more slender effective body (smaller θ). Here, the effective body is outlined by the dividing streamline along the shear layer and the physical body elsewhere. The form of the effective body dictates the pressure field in two ways namely, the inclination of the separation shock and the flow expansion ahead of separation. Firstly, the inclination of the dividing streamline governs the strength of the separation shock and, consequently, the entire pressure field downstream of separation. Hence, if the spike length is increased, the effective body becomes more slender. Secondly, the existence of an expansion fan further reduces the downstream pressure field, which is indicated by the initial pressure drop on the spike, Fig. 3a. Hence, if the spike length is increased, the separation point shifts downstream along the spike and, consequently, the extent of the expansion fan increases. Overall, spike length dictates the pressure field around the entire spiked body. Increasing the spike length yields a further reduction in the pressure over the whole body surface including the reattachment point.

The mechanisms of drag reduction using aerodisks, their relative superiority, and the further drag reduction achieved by increasing their sizes are not clear from the current explanation and the previous investigations so far. Fig. 6a illustrates the variation of local pressure along the forebody of four models having a spike length of $L/D=1.5$ and different aerodisk sizes. For comparison, the corresponding pressure distribution for a model with a pointed spike of the same length is also plotted. Fig. 6b–e shows the shapes of the corresponding effective body as outlined by the dividing streamline.

In addition to the pressure reduction as the aerodisk size increases, two other phenomena can be addressed. Firstly, in the model with $d/D=0.4$, the surface pressure is further reduced despite that the reattachment point is shifted slightly towards the spike root. Secondly, a pressure reduction is achieved on the model with the sharp-pointed spike ($d/D=0.0$) simply by rounding the spike tip ($d/D=0.05$). The hypothesis of the effective body based on the dividing streamline is employed for the explanation of drag reduction mechanism in models with aerodisks.

It can be shown that increasing the aerodisk size pushes the separation point (which is located at the aerodisk shoulder) further away from the forebody axis. Nevertheless, the reattachment point is shifted further towards the forebody shoulder. However, the overall effect of increasing the aerodisk size is that effective body becomes more slender. It can be argued that the effect of the strong bowed foreshock is local and limited to the smaller aerodisk surface and the expansion fan at the aerodisk shoulder acts to reduce the downstream pressure field. Consequently, the forebody is not affected by the strength of the foreshock. Moreover, it can be deduced that the size of the expansion fan (and hence the downstream pressure drop) increases with the aerodisk size since the dividing streamline inclination decreases at the separation point.

In the model with $d/D=0.4$, the slenderness of the effective body and the size of the expansion fan yield a further reduction in the pressure field around the entire body despite that the reattachment point is closer to the spike root. The further

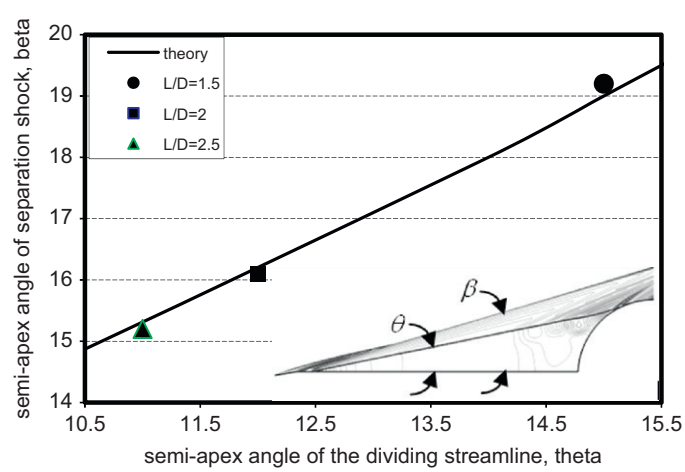

Fig. 5. Inclination of the dividing streamlines and separation shocks in models with pointed spikes.

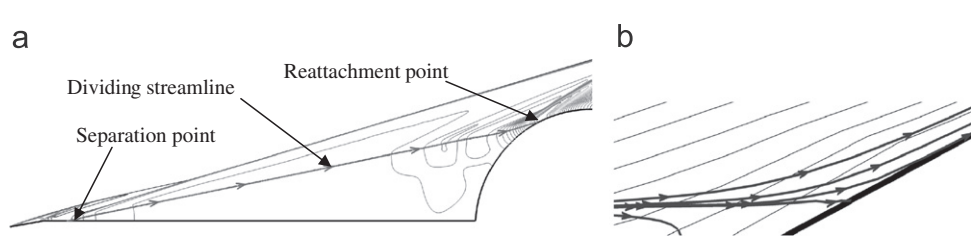

Fig. 4. Details of the dividing streamline.

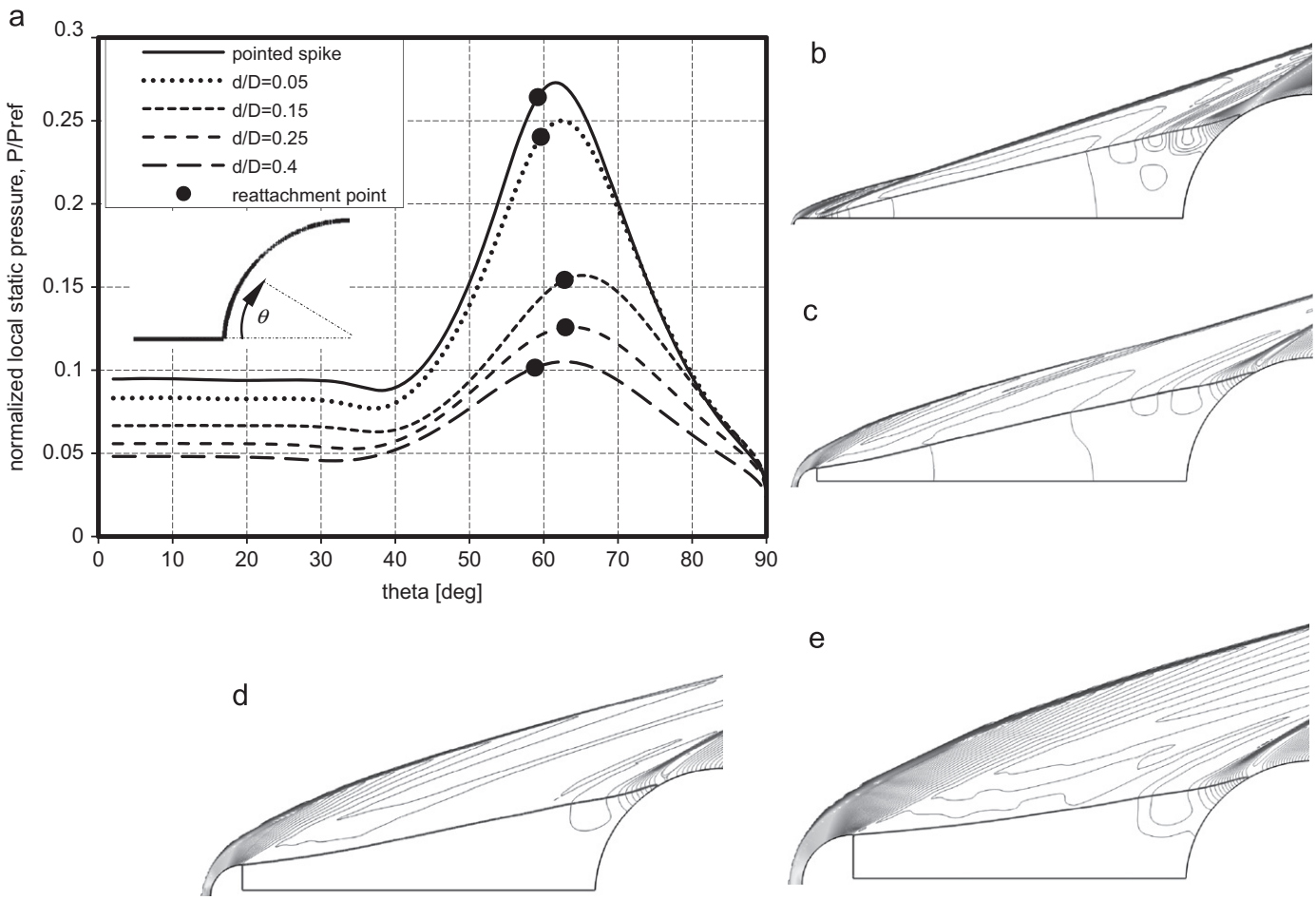

Fig. 6. Impact of the aerodisk size on the pressure field and the effective body shape.

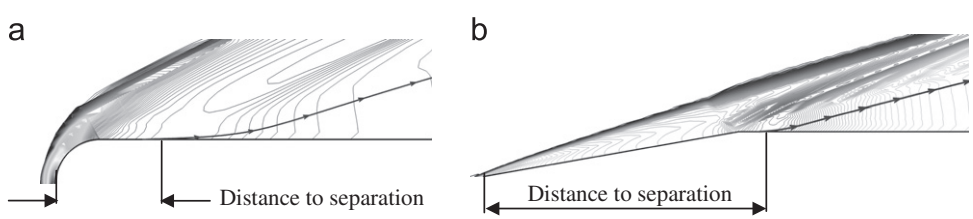

Fig. 7. Impact of rounding the spike tip on the flow details at separation: (a) rounded spike and (b) pointed spike.

pressure reduction gained by rounding the spike tip, $d/D=0.05$, compared with the sharp-pointed spike can be explained as follows. Clearly, replacing the pointed spike tip with a rounded one shifts the separation point slightly upstream. Hence the associated effective body becomes slightly more slender. To illustrate this, a close-up to the separation zone of both models is illustrated in Fig. 7. In addition to the earlier separation, a spike with a rounded tip is characterized by a larger expansion fan ahead of separation. The combined effect of larger expansion and more slender effective body eventually reduces the whole pressure field.

3.2.1. Factors governing the effective body shape

According to the hypothesis presented in this work, the effective body is outlined by the dividing streamline. In models having spikes with pointed ($d/D=0.0$) or rounded ($d/D=0.05$) tips, the dividing streamline is almost a straight line linking the shear layer separation and reattachment points. Consequently, the effective body associated with these models has a conical shape, Fig. 8a and b. In these models, the point of separation is governed by the adverse pressure gradient caused by the vicinity of the forebody.

In contrast, in the models with aerodisks ($d/D=0.05$), the shear layer separates at the aerodisk shoulder. According to the spike length and the aerodisk diameter, the shear layer may propagate directly towards the forebody or have a more curved profile. Hence, the shape of the effective body as determined by the dividing streamline varies from one model to the other, Fig. 8c and d. The factors dictating the geometry of the effective body are explained here. Chapman et al. [40] suggested that the parameter that dictates the flow pressure inside the recirculation zone and, hence, its size is the total pressure along the dividing streamline in the vicinity of reattachment zone. He argued that, for a stable reattachment, the pressure inside the recirculation zone is such that its corresponding total value is equal to the peak static reattachment pressure. The latter is, in turn, dictated by the flow turning angle at reattachment. Based on this argument,

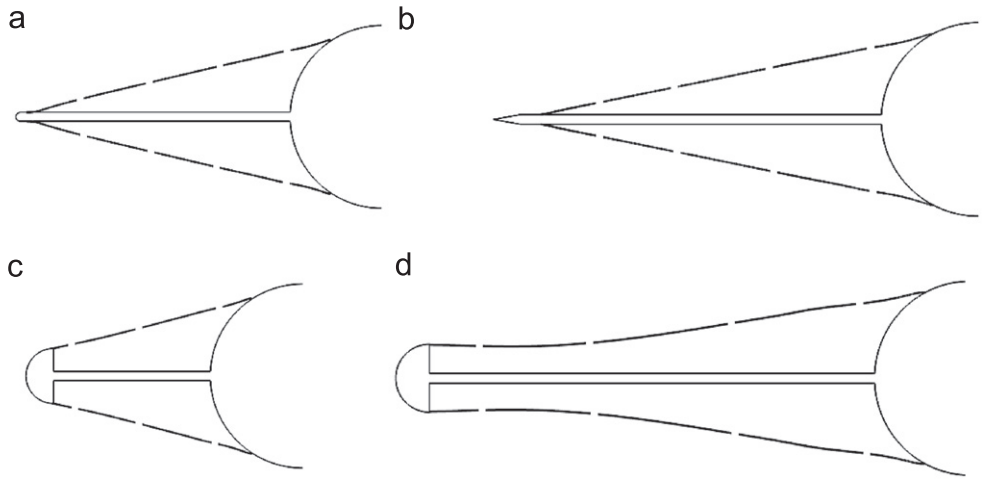

Fig. 8. Geometry of the effective body for four different models: (a) $L/D=1.5$ and $d/D=0.05$; (b) $L/D=2$ and $d/D=0$; (c) $L/D=$and $d/D=0.3$ and (d) $L/D=2.5$ and $d/D=0.3$.

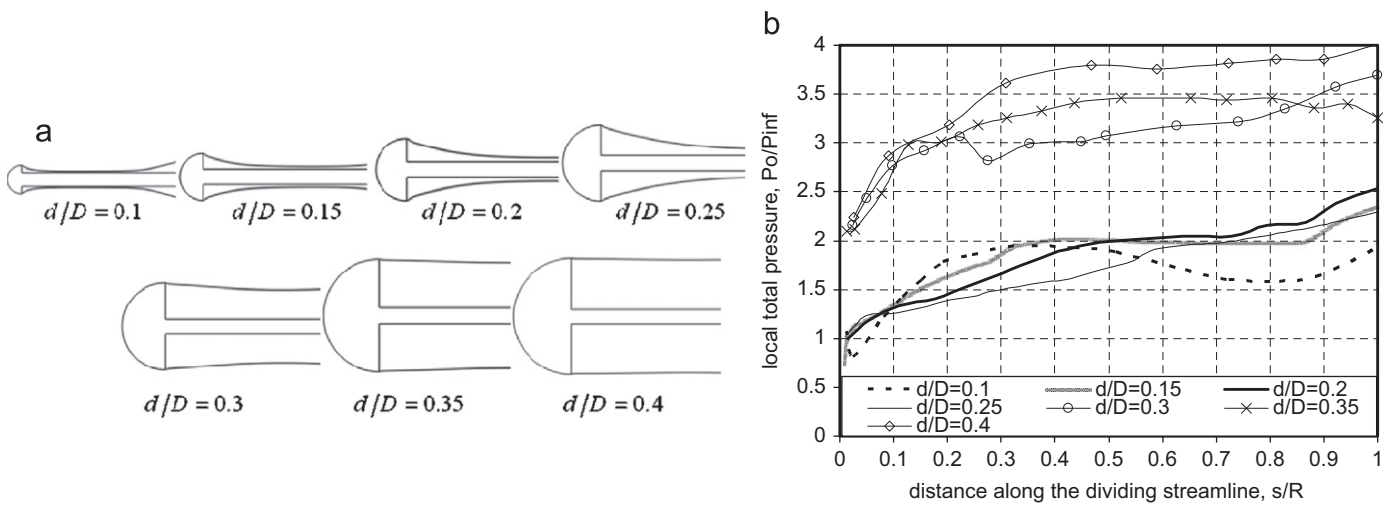

Fig. 9. Variation of (a) the shape and (b) the total pressure along the dividing streamlines downstream of separation in models with $L/D=2.5$.

Kenworthy [65] introduced the "energetic shear layer" concept as the criterion for the oscillation mode around spiked flat cylindrical models. This approach was confirmed numerically by Feszty et al. [73]. Here, an attempt is made to extend these hypotheses. The assumption is that, at equilibrium, the shape of the effective body is dictated by the energy level of the dividing streamline immediately downstream of the separation point. The dividing streamline that possesses high energy level (in terms of total pressure) immediately downstream of the separation point manages to proceed directly to the reattachment point on the forebody surface. On the other hand, the dividing streamline that has low pressure (and, consequently, low energy) gets closer to the spike and even reattaches to it. Hence, depending on the spike length and the aerodisk size, local values of Mach number and pressure attain certain values and their combination determines the energy level in the dividing streamline and eventually dictates the outlines of the effective body. To validate this assumption, the dividing streamline of the models having the same spike length ($d/D=0.05$) is investigated. Fig. 9a and b illustrates the evolution of the shape and the total pressure along the dividing streamline immediately after separation from the aerodisk shoulder, respectively.

At equilibrium, the inclination of the dividing streamline dictates the size of the expansion fan at the aerodisk shoulder and, hence, the local values of pressure at the aerodisk base. Thus, models with large aerodisks yield dividing streamlines that are less inclined with relatively higher energy levels. Hence, they separate at the aerodisk shoulder and do not reattach on the spike's surface. In contrast, the dividing streamline associated with small aerodisks start with low energy levels and they reattach on the spike's surface immediately after separation from the aerodisk shoulder. In the latter case, as the dividing streamline approaches the forebody, it experiences a pressure rise and eventually manages to separate from the spike and propagates towards the forebody. In both the cases, the total pressure of the dividing streamline continues increasing as it passes through the compression wave at the reattachment zone. According to Chapman et al. [40], if the total pressure of the dividing streamline equals the peak pressure downstream of reattachment shock, the dividing streamline stagnates on the forebody and a stable shear layer is attained.

3.2.2. Mechanism of aeroheating reduction and the aerothermal characteristics of aerodisks

Holden [30] showed that value of peak aeroheating at the reattachment point is governed by the "geometry of reattachment". The value of the peak heating increases as the angle between the reattaching shear layer and the normal to the body surface, the reattachment angle, decreases. Apart from Holden's

study [30], a comprehensive investigation of the mechanism of aeroheating reduction on spiked bodies was not conducted. The authors conducted a study [86] devoted to addressing this phenomenon on the models illustrated in the beginning of Section 3.1. The variation of wall temperature along the forebody surface of three models with sharp-pointed spikes is shown in Fig. 10. The adiabatic wall temperature is normalized with respect to the adiabatic wall temperature at the stagnation point of the unspiked body in the same freestream conditions, T_{ref}.

Consistent with Holden [30], the peak aeroheating value depends on the reattachment angle. Here, as the spike length increases, the reattachment point moves towards the forebody shoulder, and the strength of reattachment (expressed in terms of the reattachment angle) decreases. Hence, the peak temperature decreases. This reflects a major difference between pressure and temperature peaks. While the former is created by the reattachment shock wave, the latter is created due to the impingement of the shear layer on the forebody surface. Therefore, the reattachment point can be more accurately located using the peak heating rather than the peak pressure. However, increasing the spike length reduces the wall temperature ahead of the entire body and not only in the reattachment zone. The aeroheating on the forebody surface inside the recirculation zone is explained as follows. Recalling that the aeroheating is directly related to the skin friction (and hence, the velocity gradient) through the Reynolds analogy [87], it is better to first investigate the vortex structure inside the recirculation zone. The recirculation zone incorporates three vortices. The primary one is situated immediately below the shear layer and is initiated by the impingement of the latter on the body surface. A smaller secondary vortex is established below the primary one and a tiny tertiary one is created at the spike root. The primary vortex, which is the largest, increases in size as the spike length increases on the expense of the neighboring vortex. This is illustrated in Fig. 11 while Fig. 12 illustrates the variation of skin friction coefficient profile along the forebody with spike length.

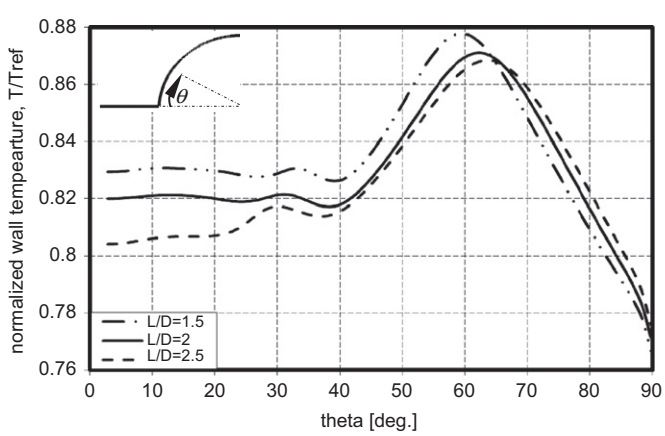

Fig. 10. Variation of forebody surface temperature with the spike length.

The flow in the primary and tertiary vortices generates a negative shear stress whereas the flow in the secondary vortex and downstream of reattachment zone generates a positive shear stress. For the short spike, $L/D=1.5$, the velocity gradient at the forebody wall caused by the primary vortex is higher due to the smaller size of the vortex. This causes the aeroheating to increase in the portion of the forebody exposed to the primary vortex. Increasing the spike length yields a more slender effective body. Hence, the velocity gradient and the skin friction of the primary vortex decrease yielding a lower surface heating to the forebody surface. The mechanism of aeroheating is different in the spiked and unspiked cases. In the unspiked forebody, the flow behind the strong bow shock wave attains significantly high temperature levels that are responsible for the aeroheating of the forebody surface. In contrast, in the spiked forebody, the aeroheating at the reattachment is caused by the impingement of the shear layer on the surface and the consequent vorticity generation. The variation of adiabatic wall temperature and shear stress on the forebody surface with the aerodisk size is illustrated in Fig. 13 for models with $L/D=1.5$. The aeroheating increases rather than decreases by replacing the pointed spike tip with a rounded one. This finding is consistent with the experimental measurements of Motoyama et al. [21]. They showed that the total heat flux to the model with a rounded-tip spike exceeded that of the sharp-pointed spike. The impact of the aerodisk size of the reattachment heat flux was not addressed.

As inferred from Fig. 13, by increasing the aerodisk size, the levels of aeroheating remain almost the same regardless of the aerodisk size; this trend can be explained as follows. Rounding the spike tip and increasing the aerodisk size increase the shear layer energy in two aspects namely, the large size of the expansion fan ahead of the separation shock and the low inclination of the latter. Consequently, the shear stress in the vicinity of the forebody increases and, hence, the aeroheating increases. Further increasing the aerodisk size causes the shear layer to be more energetic and the primary vortex to be larger. The overall outcome is an insignificant variation in the peak aeroheating at the reattachment point. This trend of forebody surface temperature variation with aerodisk size persists for models with longer spikes. This is illustrated in Fig. 14.

In all cases, spikes and aerodisks yield a reduction in the aeroheating to the entire forebody. They also reduce the peak heating value and shift its location away from the stagnation point, which is generally occupied by heat-sensitive instruments such as seekers. However, being exposed to the freestream, the spike tip and aerodisks are expected to encounter excessive aeroheating in place of the forebody. It is thus useful, in practical applications, to be aware of the aeroheating levels attained in this spot and the possible countermeasures. We can expect that aerodisks are exposed to higher aeroheating levels compared with pointed spikes. In either case, the heating intensity is high since it affects a very small surface area. A special heat-resistant material or a thermal coating (protective shields) can be used in the production of the spike tip/aerodisk.

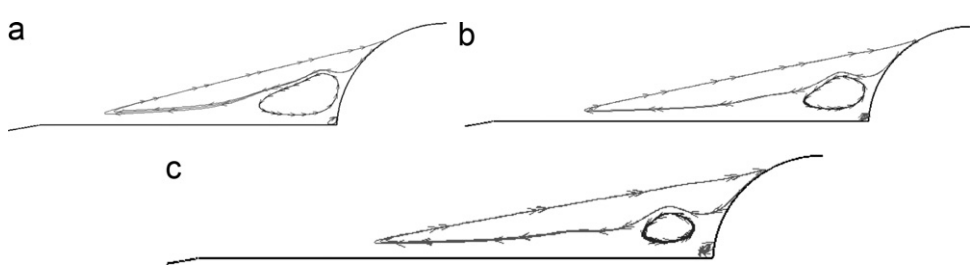

Fig. 11. Vorticity structure inside the recirculation zone for three models with sharp spike: (a) $L/D=1.5$; (b) $L/D=2$ and (c) $L/D=2.5$.

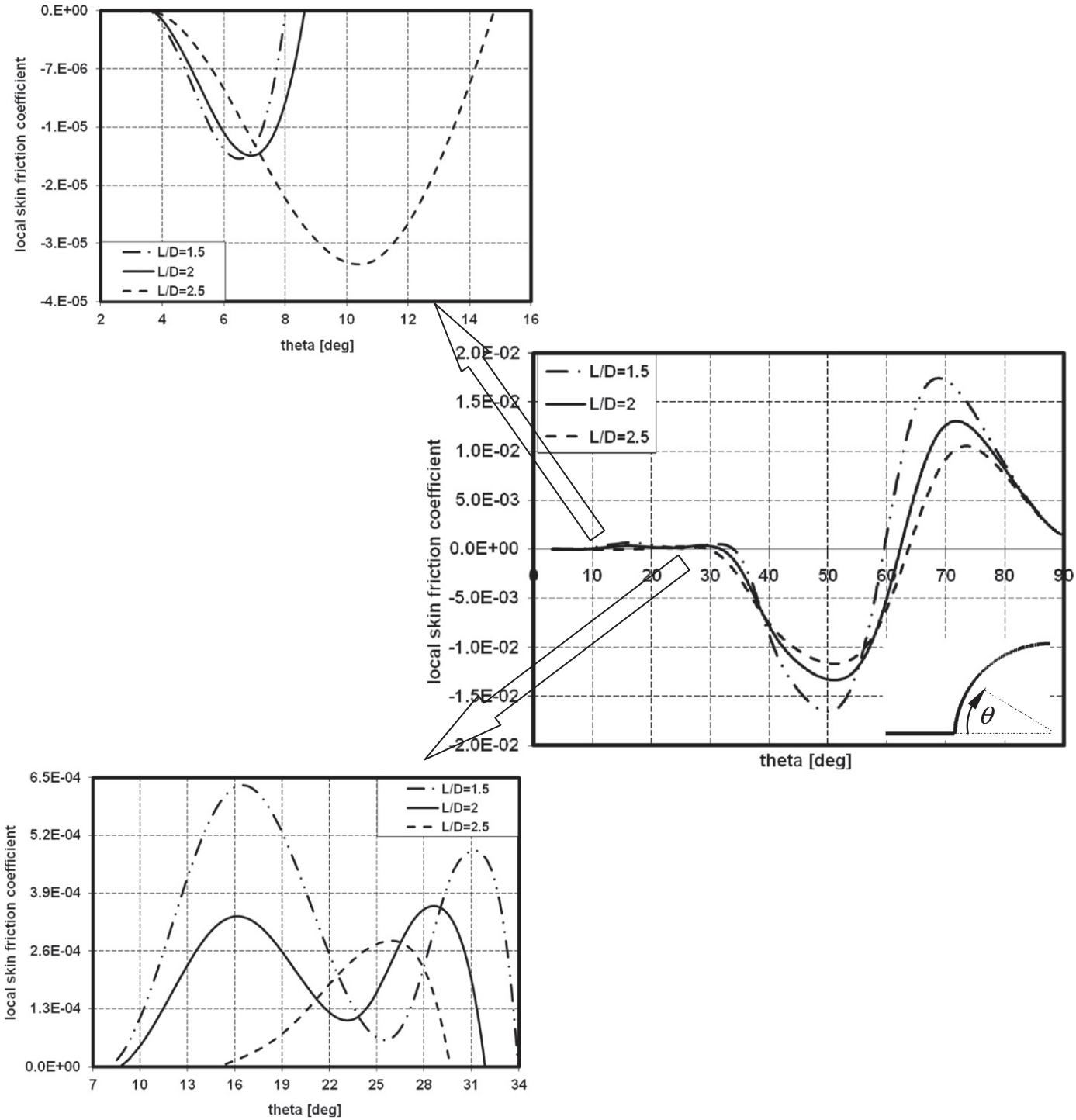

Fig. 12. Variation of skin friction coefficient along the forebody with spike length.

3.2.3. Impact of the spike and the aerodisk on the base drag

It is well established that the spikes reduce the pressure drag on the forebodies of the high-speed vehicles by reducing the local surface pressure. A more forebody pressure drag reduction is gained using aerodisks. But what is the subsequent impact of using aerodisks on the base drag? The answer to this question was not comprehensively discussed in the literature. Bogdonoff and Vas [11] argued that using a spike would reduce the base pressure. In the experimental investigations [10,16,25], the aerodisks were found to yield less total drag compared with pointed spikes. In these experiments, the drag on the whole model was directly measured using force balances and the individual drag components were not investigated.

To address this aspect, the authors conducted a special numerical experiment [81] involving two models having a hemispherical forebody and the same spike overall length of $L/D=1.5$. Model 1 is equipped with a pointed spike whereas model 2 is equipped with an aerodisk having $d/D=0.15$. The computed pressure and velocity contours for both models are illustrated in the upper and lower halves of Fig. 15a and b, respectively.

The pointed spike generates a conical attached foreshock whereas the aerodisk yields a bow detached foreshock. However,

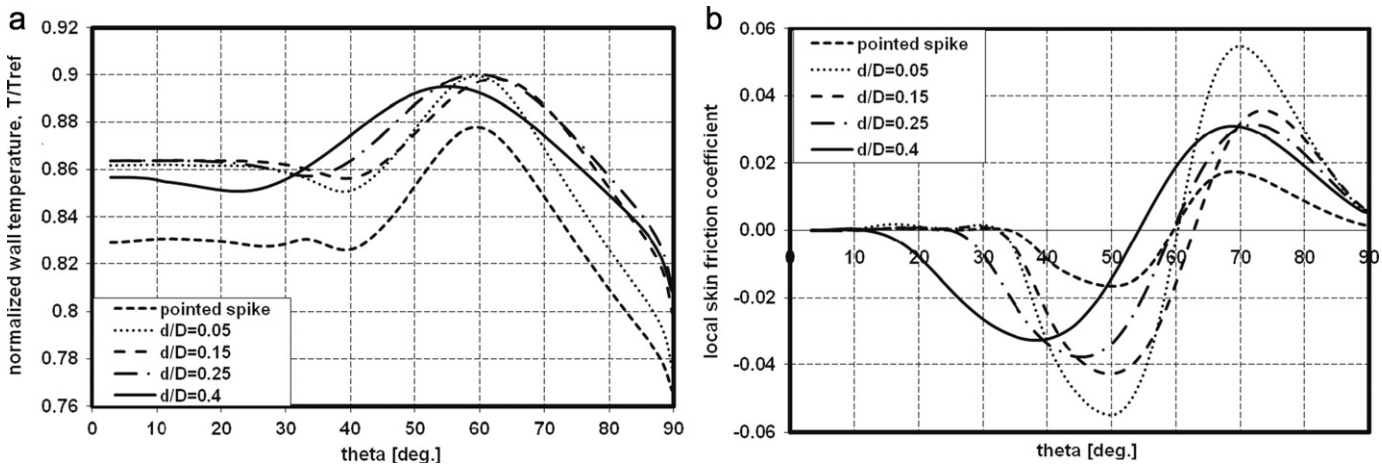

Fig. 13. Impact of aerodisk size on the forebody (a) temperature and (b) skin friction coefficient.

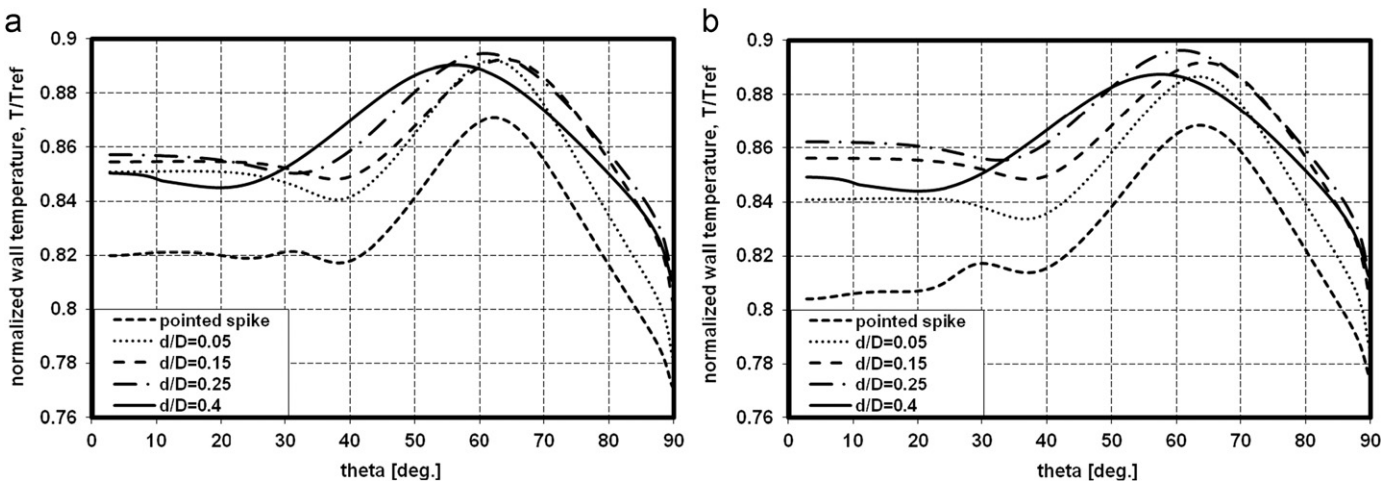

Fig. 14. Variation of forebody wall temperature profile with aerodisk size: (a) $L/D=2$ and (b) $L/D=2.5$.

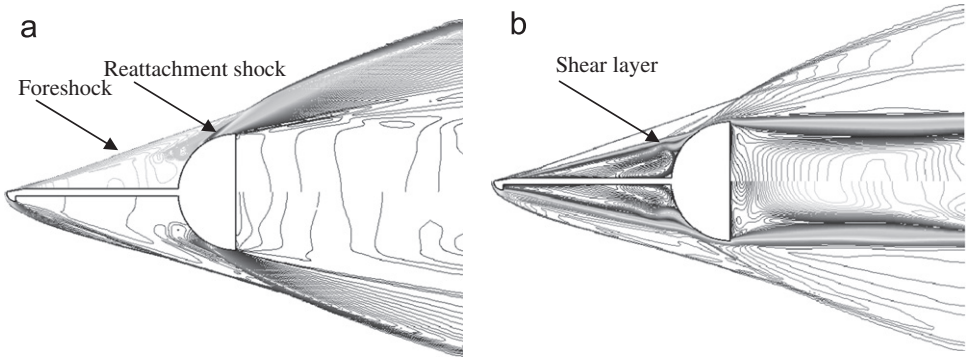

Fig. 15. Flowfield features of the extended computational domain: (a) pressure contours and (b) velocity contours.

as explained before, the flowfield pressure downstream of the foreshock is mainly dictated by the form of the effective body outlined by the dividing streamline. The effective body in case of model 2 is slightly more slender than that of model 1. In both models, the flow separates at the forebody shoulder forming a shear layer that propagates downstream of the forebody base. Downstream of the reattachment shock in both models, the flow expands dictated by the curvature of the forebody shoulder.

Since the reattachment shock in model 1 is much stronger, the local pressure at the forebody base is higher. Fig. 16a and b illustrates the pressure coefficient profiles along the forebody face and base for both models, respectively. It is clear that, compared with model 2, the pressure is higher in model 1 everywhere along the forebody face and base. This is reflected in the corresponding pressure drag coefficient as shown in Table 1.

Despite that the aerodisk yields a higher base drag, the drag reduction on the face of the model is more pronounced. The base drag on both models is two orders of magnitude less than that on the models' faces. Overall, the total drag is reduced by replacing the pointed spike with an aerodisk. We would expect that,

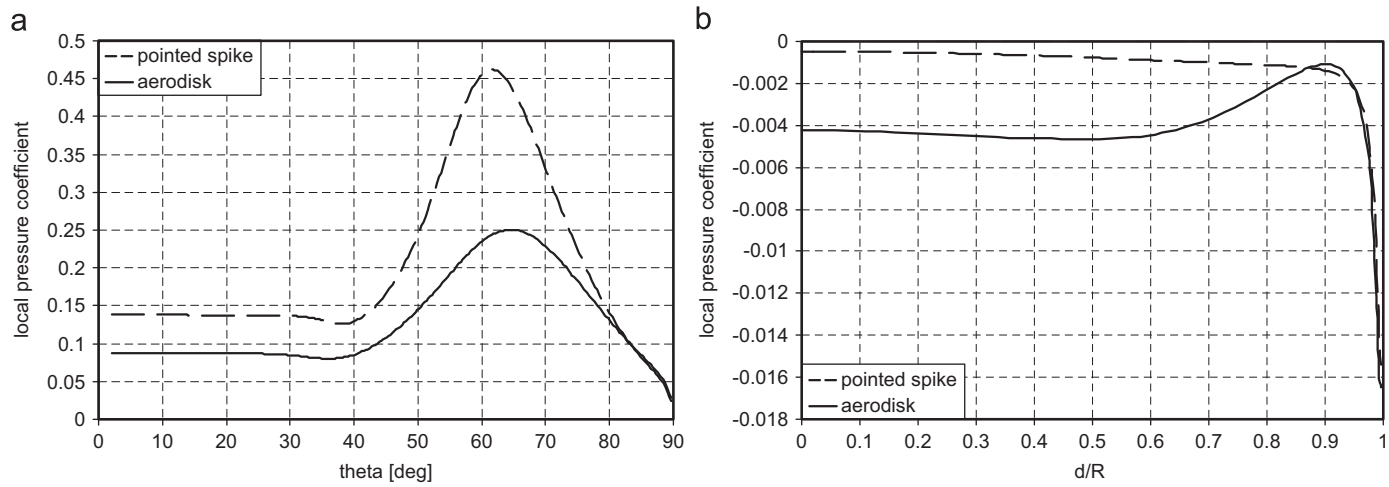

Fig. 16. Pressure coefficient profiles along the forebody (a) face and (b) base.

Table 1
Components of the pressure drag coefficients for base drag test.

	Forebody face	Forebody base	Total pressure drag coefficient
$d/D=0$	0.23164	0.00168	0.2339
$d/D=0.15$	0.1418	0.00444	0.1658

increasing the aerodisk size has the effect of increasing the base drag but reducing the overall drag.

3.2.4. Mechanism of flow instability around spiked hemispherical bodies

The flow instability associated with spiked bodies was thoroughly investigated in many studies. However, these studies mainly focused on models in which flow instability is relatively more pronounced, namely, flat cylindrical and cone cylindrical models. On the other hand, flow instability around spiked hemispherical models gained less attention. Mair [36] and Maull [9] did not record any flow instability associated with such models. However, Crawford [23] indicated signs of shear layer and foreshock "waviness" in his experiments. Yamauchi et al. [51] also reported flow unsteadiness in their own experiments at Mach 2 and 4.5 and supported their experimental findings with numerical analysis. They also reproduced numerically the experiments of Crawford [23]. In all cases, flow instability was claimed to be caused by the unsteady vortices inside the recirculation zone. In his numerical reproduction of Crawford's experiment, Mehta [52] recorded periodic pressure variation on both spike and forebody surfaces, which was argued to indicate self-sustained flow oscillations. Nevertheless, no comprehensive investigation on the occurrence of flow instability around hemisphere cylindrical bodies, its cause, mechanism, and the associated range of model designs has been conducted so far. In the context of their research [81], the authors investigated the flow stability around such bodies; the models mentioned in the beginning of Section 3.1 are investigated. The majority of these 45 models showed stable flow features. However, for some models the flow was unstable. The impact of the model design on the associated flow stability status is shown in Fig. 17.

The hollow and solid markers refer to models with stable and unstable flow modes, respectively. Generally, for the specific freestream conditions investigated, the flow around the spiked model can be stable if the spike length exceeds the forebody

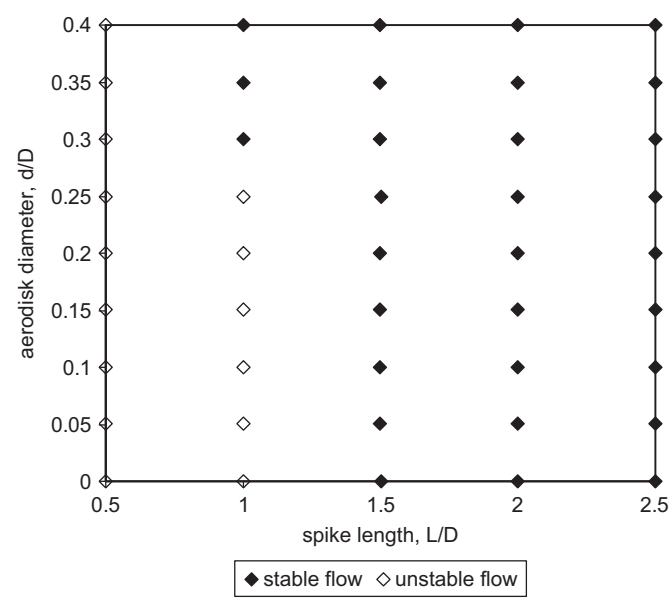

Fig. 17. Dependence of flow stability on spike length and aerodisk size.

diameter. More interestingly, increasing the aerodisk size can stabilize the flow. Models with ($L=D$) experience a stable flow if the aerodisk diameter exceeds 25% of the forebody diameter.

The flowfield of a stable flow maintains a fixed pattern that does not change with time. As suggested by Chapman et al. [40], the fundamental criterion of a stable flow is that the dividing streamline emanating at the separation point is the same dividing streamline stagnating at the reattachment point, Fig. 18a.

This only occurs if the energy level of the dividing streamline (in terms of its total pressure) at the reattachment point is sufficiently very close to peak pressure generated by the reattachment shock wave. This pressure balance on both sides of the dividing streamline at reattachment imposes a mass balance. Hence, the dividing streamline keeps a fixed position, and the shape of the effective body (Fig. 18b) remains unchanged. Eventually, the flowfield features and the associated drag remain unchanged.

On the other hand, the flowfield associated with this model continuously changes its pattern with time. The instantaneous static pressure contours around a model experiencing an unstable flow at different instances are shown in Fig. 19 (frames a–i), while their corresponding drag values are plotted in the top left of the figure.

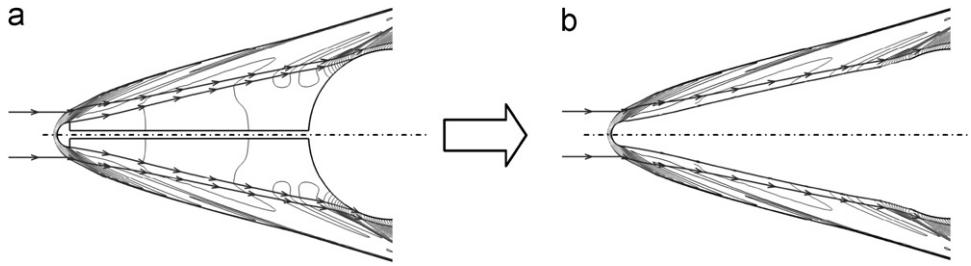

Fig. 18. Criteria of the stable flow mode.

Closely examining the frames below shows that, the streamline emanating at the aerodisk shoulder either reverses into the recirculation zone (frames a–c) or escapes downstream of the reattachment zone (frames d–h) and another streamline reattaches on the forebody surface. Since no single streamline links the separation and reattachment points, the term "dividing streamline" is no longer accurate. Feszty et al. [73] introduced the terms "bounding" streamline and "escape" streamline to refer to that emanating from the aerodisk shoulder and that reattaching on the forebody surface, respectively. However, we find it more meaningful to introduce two new terms namely, "separating" and "reattaching", respectively. Depending on the mutual position of the separating and reattaching streamlines, two different situations can be distinguished.

In frames a–c, the reattaching streamline originates from the farfield and engulfs the separating streamline. The latter then reverses into the recirculation zone. In contrast, in frames d–h, the reattaching streamline originates from within the recirculation zone and is engulfed by the separating streamline. The latter manages to escape downstream along the reattachment zone. In either situations, the location and, hence, the strength of the reattachment shock wave are dictated by the stagnation location of the reattaching streamline (reattachment point). A comprehensive illustration of the mechanism of flow instability and a complete description of its cycle can be found in [84]. They are briefly illustrated below.

The mechanism of flow instability can be explained based on the energy level of the separating streamline. If the energy level of the separating streamline is such that its total pressure as it approaches the reattachment zone is insufficient, it reverses into the recirculation zone. Another "more-energetic" streamline emanating from the freestream stagnates on the forebody at the reattachment zone. Such arrangement creates a net mass influx to the recirculation zone causing its inflation. This mass influx energizes the separating streamline such that its total pressure at reattachment gets higher than that required for reattachment. Another "less-energetic" streamline emanating from the recirculation zone stagnates on the forebody at the reattachment zone. Such arrangement creates a net mass outflux from the recirculation zone causing its collapse. This reduces the energy of the separating streamline such that its total pressure at reattachment gets lower than that required for reattachment and the cycle is repeated in a self-sustained manner.

The mechanism of flow instability as explained above is based upon the status of the energy level (in terms of total pressure) of the dividing streamline. To confirm this assumption, the evolution of total pressure along the separating streamline (S, dashed lines) and the reattaching streamline (R, solid lines) in both frames (d) and (i) are illustrated in Fig. 20. The dash-dot lines indicate the value of peak pressure (P) at the reattachment zone. Clearly, in both situations, the terminal energy level of reattaching streamline is very close to the peak pressure value. The total pressure of the separating streamline in frame (d) is higher than the peak pressure at stagnation. In contrast, the separating streamline in frame (i) possesses a total pressure, which is always below the peak pressure value.

An overview of a complete cycle of flow unsteadiness indicates that it falls below the flow oscillation mode type in which both the shear layer and the foreshock perform a lateral flapping-like motion with small amplitude and low frequency. Kenworthy [65] recorded and explained the same characteristics of oscillation flow mode on flat cylindrical bodies. Here, flow instability of the oscillation mode is extended to hemispherical models that were not studied in the previous investigations.

3.2.5. Revisiting the assumption of flow axisymmetry at zero incidence

At zero incidence, the flow around the spiked models is assumed by researchers to be axisymmetric even for models experiencing flow unsteadiness. This assumption was confirmed in a number of experimental works [6,20,51] and was adopted in all the previous numerical studies. The validity of this assumption in numerical investigations was explicitly addressed in [51,54,74,75]. Only Calarese and Hankey [18] showed that, in some cases, there could be a degree of flow asymmetry about the spike axis. In this section, the validity of flow axisymmetry assumption is revisited by highlighting the differences between the axisymmetric and the three-dimensional solutions of the same model. Here, a model having a hemispherical forebody and a pointed spike with $L/D=1.5$ is studied in a full three-dimensional domain. One quarter of the grid used in this investigation is shown in Fig. 21. The grid is clustered and aligned in the location of the shear layer and the foreshock.

Both solutions gave the same overall performance; the total drag coefficient on the model was found to have the same value. The flowfield features in both solutions are compared in Fig. 22. Both the pressure and the velocity contours are shown for the axisymmetric domain and the pitch and yaw planes of the 3D domain.

Closely examining the above figure indicates that both the axisymmetric and 3D solutions are captured nearly identical general flow features. The slopes of the foreshock and the reattachment shock and the location of their intersection are the same. The points of shear layer separation and reattachment and its inclination are also identical in both domains. However, the 3D solution indicates a degree of non-axisymmetry of the flow inside the recirculation zone. To illustrate this, the pressure and velocity contours at different locations along the spike are shown in Fig. 23.

The gauge pressure contours are clearly non-circular, which yields a circumferential rotation of the flow inside the recirculation zone as inferred from the velocity contours. The level of non-axisymmetry increases as the flow approaches the forebody face. Outside the recirculation zone, the flow is axisymmetric as indicated from the perfectly circular pressure and velocity contours. Since the flow inside the recirculation zone cannot maintain a fixed pattern, the 3D solution is switched to the time-dependent

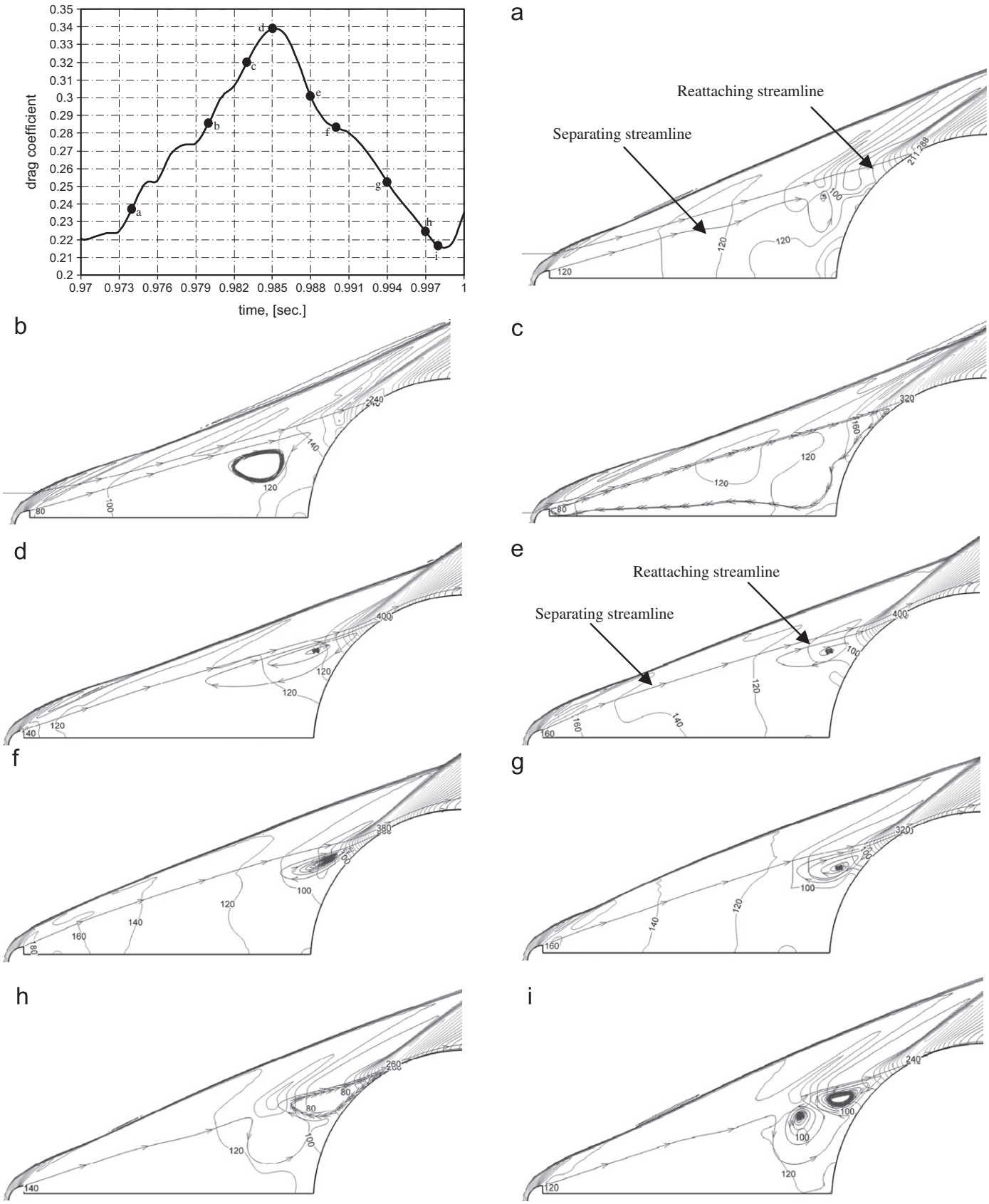

Fig. 19. Variation of flowfield and the drag coefficient in the unsteady flow mode.

solver. The total drag coefficient as predicted by the time-dependent 3D case was found to vary within 10 drag counts about the mean value. Three check points (corresponding to different time instances) were chosen to investigate the variation in the flowfield structure. Fig. 24 illustrates the instantaneous flowfield features at these points.

Apart from the recirculation zone, the flow maintains the same features that do not change with simulation time, and it also axisymmetric. Inside the recirculation zone, the flow cannot maintain an axisymmetric flow and is, hence, no longer steady. The pressure and velocity contours are non-circular and their patterns change with simulation time. However, since the values of pressure contours are generally low and the variation between contours is insignificant, the resulting drag variation is indistinguishable.

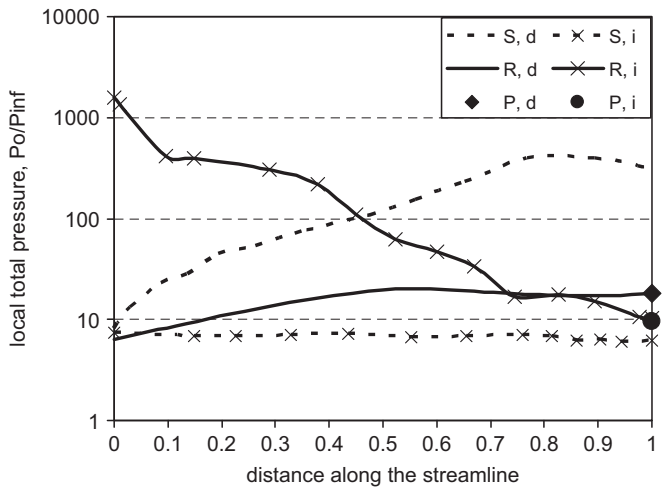

Fig. 20. Total pressure of the separating and reattaching streamlines in frames (d) and (i).

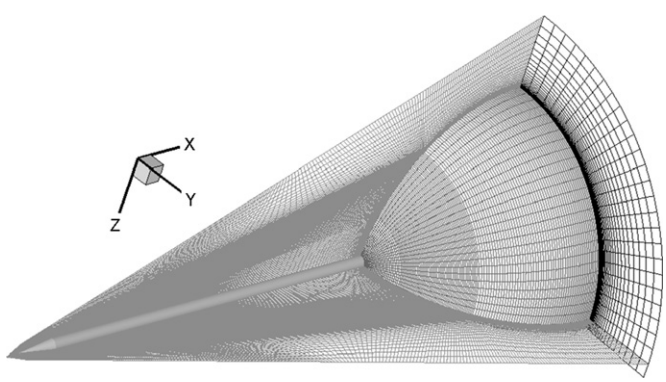

Fig. 21. One quarter of the discretized computational domain for the flow axisymmetry investigation.

3.3. Forebody design optimization of spiked hypersonic vehicles

It is well-known that, under certain flight conditions, the performance of spikes as drag and aeroheating reduction devices varies depending on both the blunt forebody geometry and the spike designs [30,61]. It is also proved that the spike's efficiency does not improve monotonically with the spike length [16,34,35,39] or the aerodisk size [21,38]. The need to optimize the spiked blunt forebody design was explicitly highlighted in the literature [4,16,21,30,38,56]. However, a study dedicated for the design optimization of spiked blunt bodies in high speed regimes was not conducted so far. The study of the authors [88] is dedicated to finding the optimum design(s) of a spiked blunt forebody through a surrogate-assisted, CFD-based, multi-objective, evolutionary optimization. The shape and geometric entities of the spiked forebody to be optimized are illustrated in Fig. 25 whereas the freestream conditions correspond to a flight at an altitude of 60 km with Mach 6 and zero incidence.

Out of the nine geometric entities illustrated in Fig. 25, six are chosen as the important, independent design parameters to be optimized; the bounds of variation of their values are as follows (per unit forebody diameter):

L=0.0:1.0, r=0.02:0.5, L_S=0.04:4.0, L_d=0.0:0.15, r_d=0.02:1.0, and r_n=0.0:0.1.

Other geometric entities are given fixed values as follows:

R=0.5, B=0.25, and r_S=0.02

The optimization problem involves two objectives namely, minimum drag and minimum aeroheating. The total drag coefficient (referred to the freestream conditions and the base area of the forebody) is taken as the measure for the drag objective. The peak heat transfer rate to the forebody is taken as a metric for the aeroheating objective. The stability of the flow around the spiked forebody is introduced as a constraint for the optimization. The amplitude of variation in the drag coefficient with numerical solution iterations is taken as the metric for the stability constraint. The values of these two objectives and the constraint are taken from the output of the CFD solver. The NSGA-II [89] is taken as the optimizer for this multi-objective constrained problem; the NSGA-II toolbox [90] in Matlab [91] is utilized in this study. Instead of directly coupling the optimizer with the "expensive" flow solver, surrogates are used. To construct the surrogates, a set

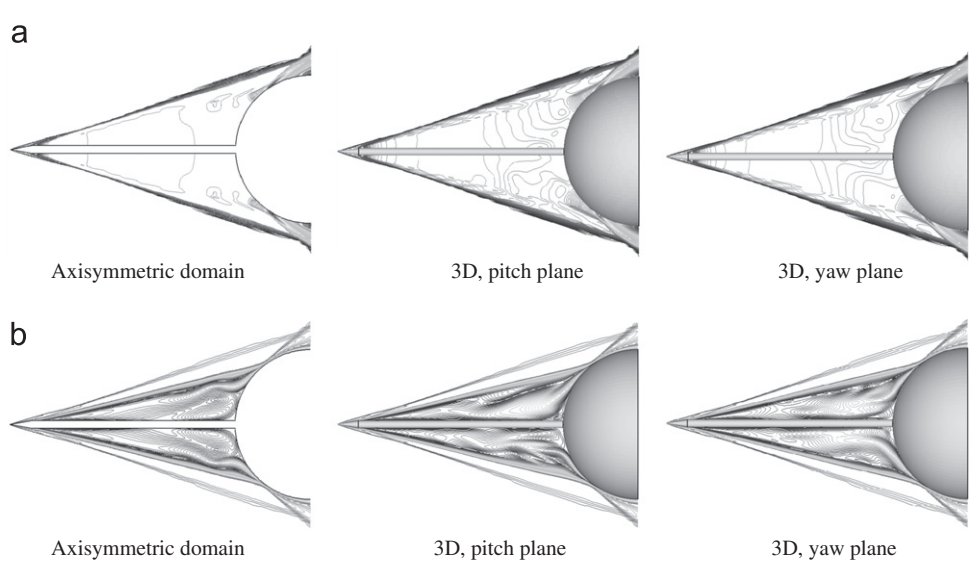

Fig. 22. Comparing the flowfield features of the axisymmetric and 3D solutions: (a) pressure contours and (b) velocity contours.

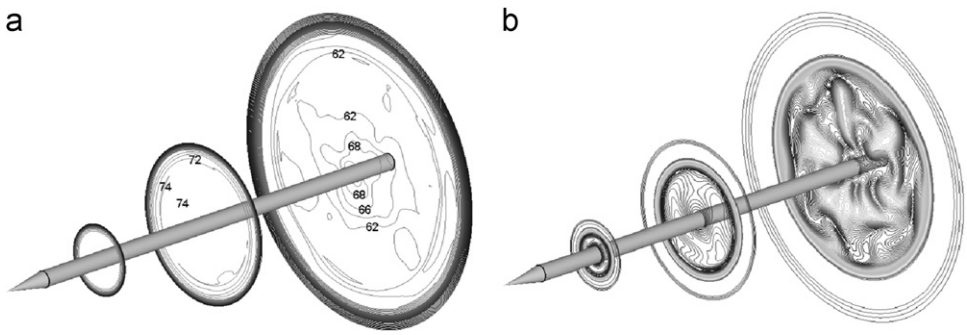

Fig. 23. Contours of the (a) pressure and (b) velocity at different axial locations.

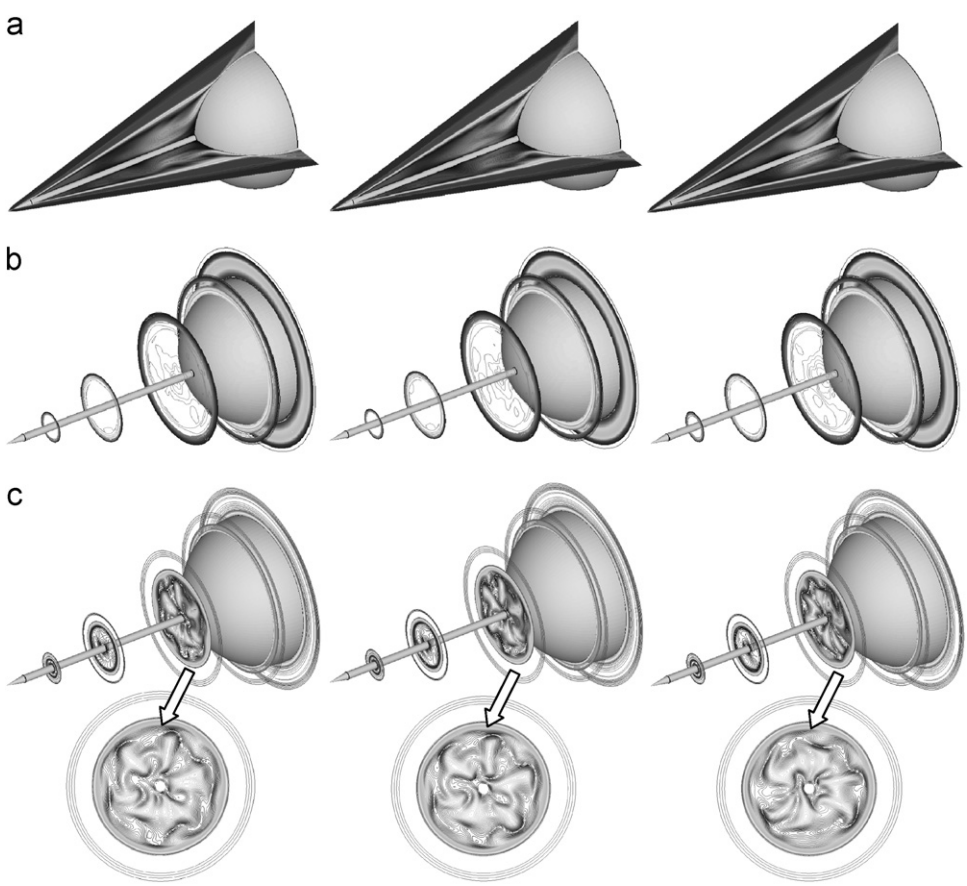

Fig. 24. The instantaneous flowfield features at different simulation time instances:(a) velocity contours on the pitch and yaw planes; (b) pressure contours on different axial locations; (c) velocity contours at different axial locations.

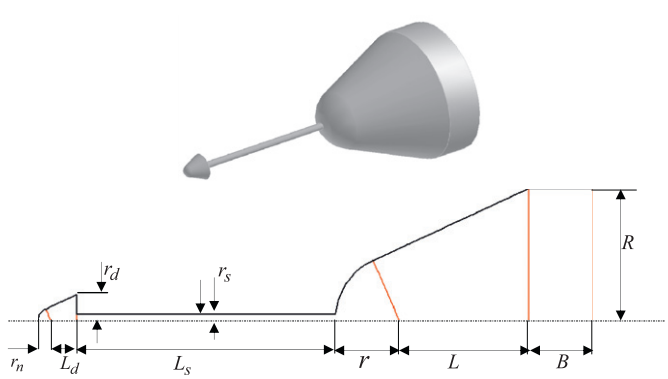

Fig. 25. Geometry and design parameters of the spiked blunt forebody to be optimized.

of 200 training samples is selected from the design space using the space-filling MM-LHD [92] sampling technique. Prior to sampling, the design parameters are scaled linearly to the same range namely, [0.1] to eliminate the effect of different dimensions. In effect, the design space can be viewed as a 6-dimensional unit hypercube with the scaled design parameters on its coordinates. Each sample in the training set resembles a unique design. The values of the objectives and constraint functions for each design are taken from the output of the CFD solution. Based on these values, a Kriging surrogate [93] is used to construct a surrogate for each of the objectives and the constraint. The SURROGATES toolbox [94] in Matlab is utilized in this study. The accuracy of the constructed surrogates is assessed using the cross-validation technique [95] and is iteratively improved by adding the so-obtained optimum designs in each iteration to the

training samples of the next iteration. Prior to optimization, a sensitivity check is performed using the Sobol's technique [96]. The values of Sobol's total indices of the six design parameters for the objective and constraint functions are shown in Fig. 26. The index value of a given design parameter reflects the degree of importance of this design parameter with respect to the response value.

Two optimization problems were considered namely, unconstrained and constrained optimization problems. In the first one, the stability constraint is overlooked whereas in the second one the stability constraint is considered. The impact of population size and generations number is tested and the solution is found to converge to the Pareto surface after 1000 generations using 1000 individuals. The locations of the Pareto optimal solutions in the objective space, the Pareto fronts, for both optimization problems are illustrated in Fig. 27.

The solid and dashed lines indicate the Pareto front for the constrained and unconstrained optimization problems, respectively. Compared with the original training samples, the Pareto optimal designs of both problems show a considerable performance improvement in both objectives. Here, the focus is on a number of key samples from the Pareto set. Models A and B lie on the aeroheating minimum of unconstrained and constrained Pareto fronts, respectively. The flow features of these models are illustrated in Fig. 28a and b, respectively. A close-up at the forebody nose of model A is shown in Fig. 28c.

The flow in both models shows almost the same features. The blunt aerodisk generates a detached bow shock ahead of it. The boundary layer separates at the aerodisk shoulder, reattaches on the spike, and remains attached. Immediately ahead of the forebody the shear layer separates and reattaches on the forebody forming a small recirculation zone. The flow details at the reattachment are shown in Fig. 28c where the arrowed line indicates the dividing streamline. According to Holden [30], the aeroheating at reattachment is mainly governed by the "reattachment geometry".

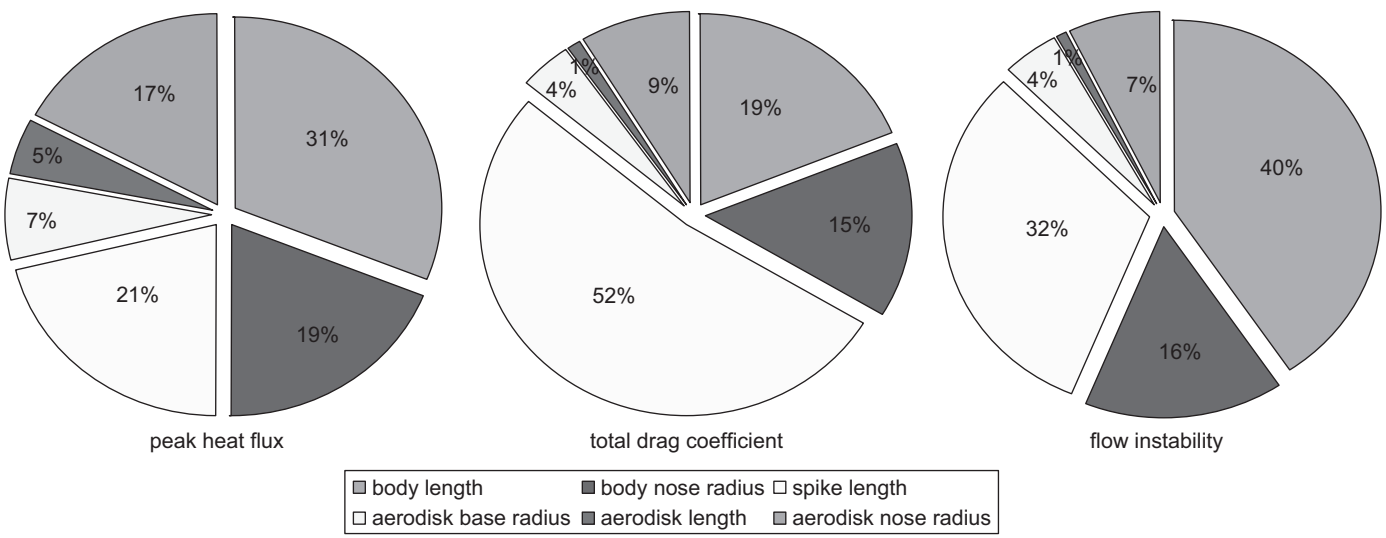

Fig. 26. Sobol's total sensitivity indices of the design parameters.

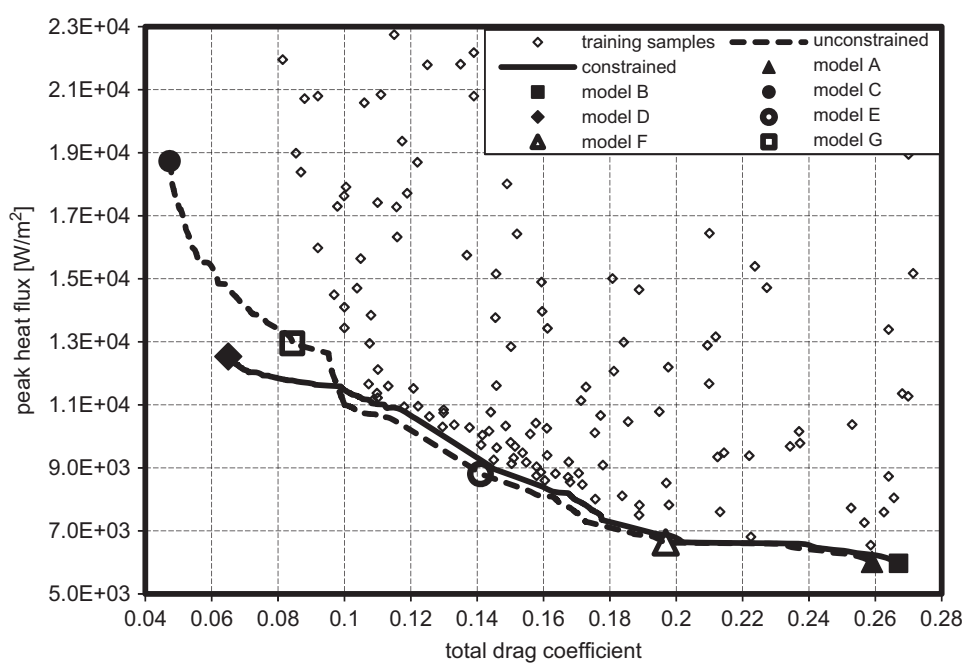

Fig. 27. Pareto front of the full-scale optimization problem.

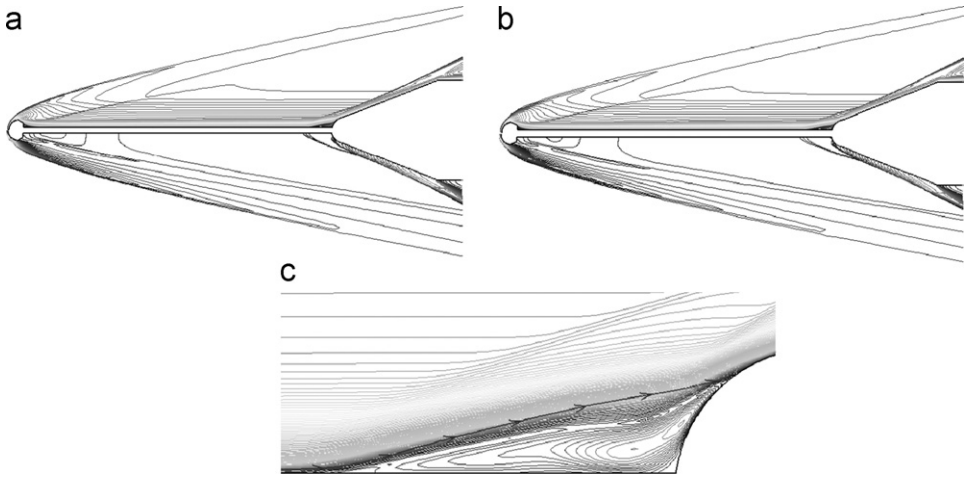

Fig. 28. Features of the numerical flowfield around (a) model A and (b) model B.

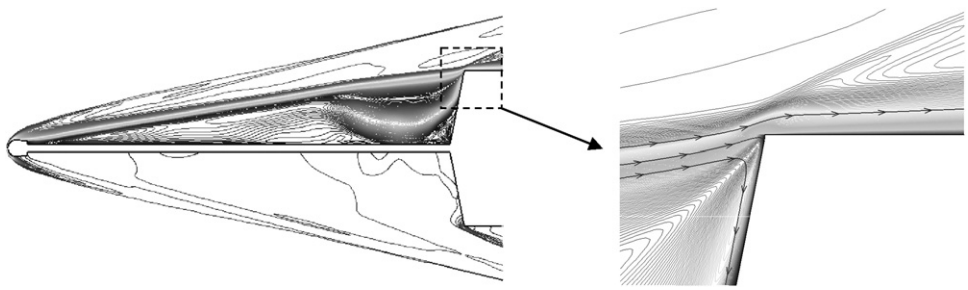

Fig. 29. Features of the numerical flowfield around model C.

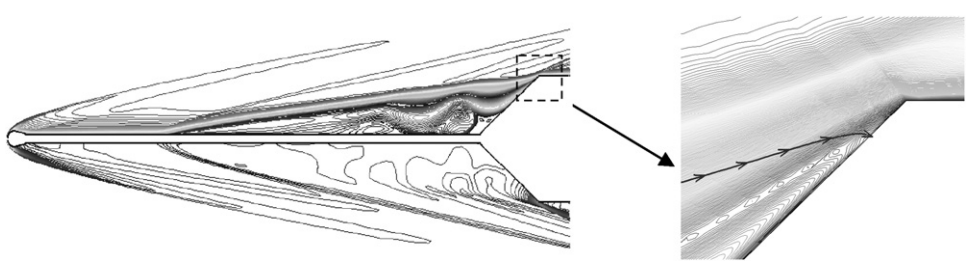

Fig. 30. Features of the numerical flowfield around model D.

In these models, the "smooth" reattachment of the shear layer on the forebody nose explains the low value of peak heat flux. However, as the flow downstream of the spike approaches the forebody, two successive oblique shock waves are created at the shear layer separation and its reattachment. The high pressure values attained by the flow downstream of these two shock waves along with a small area of the forebody being screened by the small low-pressure recirculation zone are responsible for the rise in the drag coefficient on these models.

Model C lies on the lower drag extreme of the unconstrained Pareto front. The computed velocity and pressure contours of this model are plotted in the upper and lower halves of Fig. 29, respectively. A bow shock wave is generated ahead of the blunt aerodisk. The boundary layer separates early from the spike under the influence of the highly inclined forebody face. At the forebody shoulder, the shear layer performs a partial reattachment. A partial reattachment means that part of the streamlines constituting the shear layer reverses into the recirculation zone, and others escape past the forebody shoulder, whereas the dividing stream line always stagnates at the forebody shoulder as illustrated by Wood [22]. The flow details in this area are better illustrated in the close-up on the right of Fig. 29. The resulting large recirculation zone screens the entire area of the forebody face causing the surface pressure and, consequently, the drag to become significantly low. However, as indicated by Wood [22], this arrangement of shear layer reattachment invokes flow instability in the form of lateral oscillations of the shear layer. On the other hand, the strong reattachment of the shear layer on the nearly-flat forebody face causes the peak aeroheating flux to attain a relatively high value.

Model D is located on the lower drag extreme of the stability-constrained Pareto front. The numerical flowfield features are illustrated in Fig. 30. A bow shock is generated at the aerodisk blunt tip. The shear layer separates well ahead of the forebody and reattaches partially at its shoulder. The close-up at the forebody shoulder indicates that the reattachment point is well below the acute corner. This face reattachment encourages the stability of the reattachment and, hence, a stable flow mode is established.

Model E is a sample of the intermediate models lying between the Pareto extremes. The computed velocity and pressure contours are illustrated in the upper and lower halves of Fig. 31,

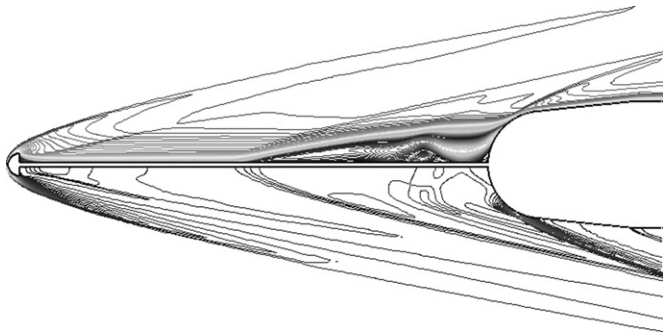

Fig. 31. Features of the numerical flowfield around model E.

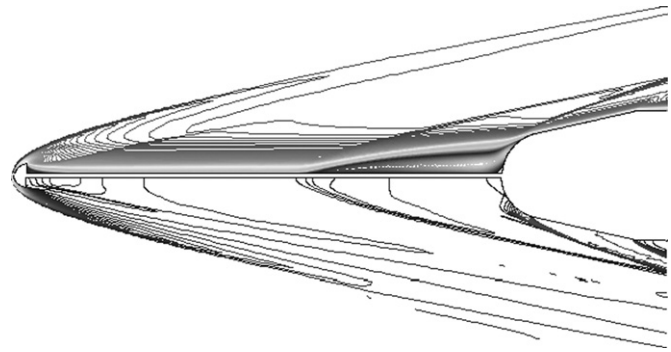

Fig. 32. Features of the numerical flowfield of model F.

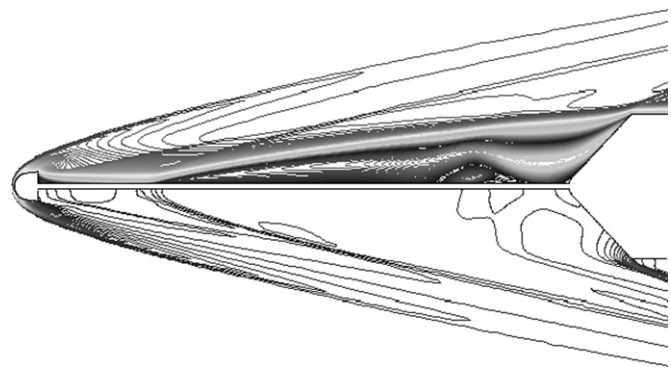

Fig. 33. Features of the numerical flowfield of model G.

respectively. A bow shock is created at the blunt aerodisk and a small recirculation zone is engulfed at its base by the shear layer that separates at the aerodisk shoulder. As the shear layer reattaches on the spike downstream of the aerodisk, a small internal shock wave is created. The bluntness of the forebody encourages the early separation of the shear layer from the spike and the formation of a relatively large recirculation zone. The long spike decelerates the flow velocity inside the shear layer and the curvature at the reattachment zone yields a smooth reattachment. Both factors contribute to the low peak heat flux value on this model. The relatively low drag on this model can be explained by two factors. On one hand, a considerable portion of the forebody face is screened by the large low-pressure recirculation zone. On the other hand, the low inclination of the forebody generates a weak reattachment shock wave, which yields a low flow pressure at and downstream of reattachment.

As the optimal design evolves from model E towards models A and B, the slenderness of the forebody (its length) increases whereas its bluntness decreases. To illustrate this, model F is selected from the Pareto front between E and A. The flowfield features of this model are illustrated in Fig. 32. The design of Model F is similar to that of model E. However, the forebody of the former is longer and its nose is less blunt. This causes the separation of the boundary layer to be closer to the forebody nose. Hence, the inclination of the effective body is higher compared with that of model E. Consequently, the flow remains attached to the spike for a longer distance. These two factors explain the rise in drag in model F compared with model E. On the other hand, the shear layer reattachment on the forebody nose is stronger in model E. This can be clearly addressed by comparing the strength of the reattachment shocks in both models. As a consequence, the Aeroheating in model F is lower than that in model E. Model F can be viewed as an intermediate design between models A and E. Increasing the slenderness of the forebody and reducing its bluntness shift the optimal designs towards the minimum aeroheating extreme, model A. Conversely, increasing the forebody bluntness and reducing its slenderness shift the designs towards lower drag and higher heating values.

On the other hand, as the optimal designs evolve from model E towards models C and D, they can have different forms. Model G represents a typical optimal design in the Pareto front from E to C. Its flowfield features are illustrated in Fig. 33. Model G has a different design compared with model E; its forebody is significantly shorter and less blunt. It can be assumed that further drag reduction by increasing the forebody nose bluntness may not yield optimal designs. Hence, different designs similar to model G appear on the Pareto front. Model G has almost the same forebody face inclination as model D. The slightly higher drag of model G can be explained by the larger aerodisk base drag. The flow expansion at this area may be responsible for the drop in the downstream flow pressure. As a consequence, the flow separates earlier compared with model D. This may cause the shear layer reattachment to be smoother, which explains the drop in peak heat flux compared with model D.

As the optimal designs evolve towards model D, the forebody becomes more inclined with less blunt nose to enhance the early separation and yield more slender effective body (outlined by the shear layer). In addition, spikes can become slightly shorter to reduce the friction drag. However, highly inclined faces of the forebody can result in an unstable reattachment, which yields unstable flowfield and hence unstable performance. The optimal models of such designs are situated towards model C on the unconstrained Pareto front. Inspired by Maull [9], we might expect that, adding a small roundness at the sharp corner of the forebody shoulder would encourage a more stable reattachment. The effect of such alternation on the optimality of these designs may invoke further investigation.

4. Conclusions and future perspective

Since their introduction to the design of high-speed vehicles in 1947, spikes and aerodisks have gained much increasing interest from the researchers. The naive modification of the vehicle's design by adding a thin rod at the vehicle's stagnation point yields a flowfield full of interesting details including, flow separation, shock–shock, and shock–layer interactions. The present paper is intended to give the readers the up-to-date advances in the field of spiked bodies aerothermodynamics. The previous literatures are thoroughly explored and the studies that founded our today's knowledge in this field are highlighted. The recent contributions of the authors are discussed and their main results are illustrated. There are many studies that are yet to be conducted in the field of spiked bodies aerothermodynamics. To name a few:

– The concept of the pivoting spike [77] seems promising and invokes more studies.

- The effect of aerodisk rotation [78–80] deserves more focus.
- The bending and vibration of spikes in practical applications [14] are not thoroughly studied.
- Wood [22] showed that rounding the shoulder of flat-faced spiked models could stabilize the flow. This aspect needs further investigations.
- Comparing the effectiveness of spikes and the similar drag reduction devices, especially forward-facing jets, may be done. A hybrid device can then be proposed.
- Finally, design optimization of spiked bodies at different flight regimes can be conducted. The design optimization can be further extended to include other design considerations such as volumetric requirements, spike bending, and vibration. These can be added either as objectives, constraints, or even in a multidisciplinary optimization.

Appendix

See Table A1 for more details.

Table A1
Summary of the previous studies of spiked bodies aerothermodynamics.

Ref.	Type of study	Conditions		Geometry			Focus of study						Special features
		M	Alpha	Forebody	spike tip/ aerodisk	*L/D*	*S*	*D*	*L*	*P*	*A*	*U*	
13	**F**	0.95:1.37	0	**BO/SO**	**CA**	1.5		▲					First study ever
35	**E**	1.76:1.93	0	**H**	**PT**	1:12		▲					
36	**E**	1.96	0	**FC/H**	**PT**	Up to 6	▲	▲				▲	First record of pulsation
37	**E**	2.72	0	**SK**	**PT**	Up to 4.5	▲						
28	**F**	0.7:1.3	0	**H**	**PT**	Up to 2		▲					The term "spike" is introduced
34	**E**	0.12:5.04	0	**H**	**FT/ CA**	0.52					▲		First aeroheating study
38	**E**	2.72	0	**H/BK**	**CA**	Up to 2		▲					Impact of spike–aerodisk length
33	**E**	1.5:1.8	0	**FC**	**PT**	Up to 4.7	▲					▲	First record of hysteresis and oscillation
39	**E**	3.5	0	**FC**	**PT**	Up to 2					▲	▲	
11	**E**	14	0	**FC/H**	**PT**	Up to 8		▲			▲	▲	
23	**E**	6.8	0	**H**	**PT**	Up to 4		▲			▲		
9	**E**	6.8	0	**FC**	**PT**	Up to 3	▲					▲	Impact of shoulder roundness
22	**E**	10	0	**SK**	**PT**	Up to 5	▲					▲	Impact of cone angle, declare flow patterns
8	**E**	19.4	0	**BO/H**	**PT**	Up to 5		▲			▲		
76	**E**	11.76	0:7	**H**	**PT**	4		▲					The spike is deflected w.r.t. model axis
29	**E**	10.1	0:25	**FC/BK**	**PT**	Up to 5		▲					Very low Reynolds number
30	**E**	10 and 15	0	**H/FC/SK**	**PT**	Up to 4	▲				▲	▲	Declare more flow patterns
26	**E**	3, 4, 6	2, 5	**BK**	**CA**	0.055: 0.083	▲					▲	Impact of very short spikes
42	**E**	9.6	0	**BK,SK**	**CA**	Up to 0.13	▲				▲	▲	
60	**E**	NA											New terms: "pulsation" and "oscillation"
61	**E**	6	0	**FC**	**PT**	Up to 3	▲	▲			▲	▲	Range of spike length for each flow modes
64	**E**	6	0	**FC**	**PT**	Up to 3	▲					▲	Explain the pulsation mode
14	**E**	Up to 3.5	>0	**BO**	**BA**	Telescoping	▲	▲				▲	Aerodisk superiority, spike bending
20	**E**	Up to 4	0.4.8	**BO**	**BA**	0.8	▲					▲	
62	**E**	2:6	0	**SK**	**PT**	Up to 5	▲					▲	Amplitude and frequency of unsteadiness
67	**E**	–	0	**FC**	**PT**	–	▲					▲	Explain the pulsation (independent of [64])
65	**E**	NA					▲					▲	Explain the oscillation mode
63	**E**	2:6	0	**SK**	**PT**	Up to 5	▲					▲	
70	**NE**	3	0	**TK**	**RT**	0.75	▲					▲	First numerical study of pulsation
66	**E**	–	0	**FC**	**PT**	–	▲					▲	
46	**N**	0.5:3	0	**FC**	**PT**	Up to 3.25	▲	▲					First comprehensive numerical study
71	**NE**	3	0	**TK**	**RT**	0.75	▲					▲	
55	**E**	2.2	0	**BO**	**FA**	1 and 1.875	▲	▲					Injection/ burning of gas jet at aerodisk
47	**N**	2: 6	0	**FC/SK**	**PT**	Up to 1		▲			▲		Low Reynolds number
18	**E**	3	0	**TK**	**RT**	0.75	▲					▲	Extensible spike
41	**E**	3	0	**H**	**CA/ PA**	0.283: 1.78					▲▲		
48	**N**	3:8	0	**TK**	**RT**	Up to 1	▲						
2	**NE**	2.5	0	**BK/H**	**BCA**	2:9	▲	▲					
27	**NE**	1.72	0	**FC**	**FT**	1.08 and 1.47	▲	▲					
68	**E**	2 and 3	0	**FC**	**PT**	Up to 1.5	▲					▲	
78	**E**	3	0, 10	**FC/SK/H**	**WA**	Up to 1.68		▲			▲		Introduce the "rotating aerodisk"
3	**NE**	4	0	**H/HC**	**FT**	0.875 and 1.3		▲			▲		
6	**E**	6.06	0: 40	**HC**	**FA**	3		▲			▲		
51	**NE,V** [23]	2.01, 4.15, 6.8	0.10	**H**	**PT**	0.5, 1 and 2	▲	▲				▲	
49	**NE**	1.72	0	**FC**	**FT**	1.08 and 1.47	▲	▲					
50	**NE**	3	0, 2	**H/FC**	**FT**	1.42	▲						
79	**E**	3	0, 10	**FC/SK/H**	**WA**	Up to 1.68		▲			▲		Rotating aerodisk
80	**E**	3	0, 10	**FC/SK/H**	**WA**	Up to 1.68		▲			▲		Rotating aerodisk
5	**NV** [23,51]	Numerical reproduction of [23,51]					▲	▲			▲		
7	**NV** [23,51]	Numerical reproduction of [23,51]					▲				▲		
74	**NV** [65]	6	0	**FC**	**PT**	1	▲					▲	
75	**NV** [65]	2.21	0	**FC**	**PT**	1.25:2.4	▲					▲	Fine numerical study of hysteresis
21	**E**	7	0: 8	**H**	**FT/RT/HA/FA**	0.5 and 1		▲		▲	▲		Drag decomposition, pitching moment
25	**E**	1.89	−4:10	**H**	**FT/RT/PT**	1		▲	▲				Impact of spike tip shape

Table A1 (*continued*)

Ref.	Type of study	Conditions		Geometry			Focus of study						Special features
		M	Alpha	Forebody	spike tip/ aerodisk	*L/D*	*S*	*D*	*L*	*P*	*A*	*U*	
32	**E**	0.92, 1.62	0	**H**	**CA**	0.773		▲					Aerodisk on struts, no stem
44	**E**	1.89	−4:10	**H**	**FT/RT/PT**	1		▲	▲				
52	**NV** [23]	Numerical reproduction of [23]						▲				▲	
1	**EN**	5.75	0:12	**BK**	**FA**	1		▲					
57	**EN**	4.5	0: 24	**H**	**FA/HA/BCA**	1	▲	▲					
56	**EN**	5.75	0	**BK**	**FA**	1	▲					▲	
10	**EN**	5.75	0:12	**BK**	**FT/HA/FA**	Up to 1		▲			▲		
4	**EN**	4.5	0: 24	**H**	**FA/HA/BCA**	1	▲	▲					
53	**NV** [23,25]	4:8	0:12	**H**	**PT**	0.5.1	▲	▲	▲	▲			
54	**EN**	2.2	0	**BO**	**FA**	1 and 1.875	▲	▲					
72	**NV** [65]	2.21	0	**FC**	**PT**	1	▲					▲	Fine numerical study of pulsation
73	**NV** [65]	6	0	**FC**	**PT**	2	▲					▲	Fine numerical study of oscillation
24	**NV** [4,6]	6.06	0:40	**HC**	**FA**	3		▲			▲		
58	**EN**	8	0	**BK**	**FA**	1		▲					
69	**E**	6.08	0	**FC**	**PT**	Up to 1.5	▲					▲	
16	**E**	6	0:8	**H**	**HA/ FA**	1.5 and 2		▲	▲	▲			
15	**NV** [23]	5, 7, 10	0	**BK**	**RT/ HA**	1:4	▲	▲			▲		
77	**E**	2, 3, 5	0:30	**H**	**PT**	-		▲			▲		"Pivoting spike" concept
31	**E**	6	0, 4	**H**	**RT**	1	▲	▲					Injection/burning of gas jet at aerodisk
12	**NV** [16]	Numerical reproduction of [16]											
59	**NV** [4,6,24]	Numerical reproduction of [4,6,24]					▲						Comparing the commercial CFD codes
17	**NV** [18,23,65,67]	Numerical reproduction of [18,23,65,67]					▲						
43	**E**	8.2	0	**H/SK**	**PT**	Up to 2.125	▲	▲					

Key for: type of study	Key for: forebody geometry	Key for: spike tip/aerodisk geometry	Key for: focus of study
E=experimental study	**BK**=blunt cone	**BA**=blunt aerodisk	**A**=aeroheating
EN=experimental study with numerical validation	**BO**=blunt ogive	**CA**=conical aerodisk	**D**=drag
F=firing test	**FC**=flat-faced cylinder,	**FA**=flat aerodisk,	**L**=lift
N=numerical study with no experimental validation	**H**=hemisphere	**FT**=flat tip	**P**=pitching moment
NE=numerical study, validation with own experimental results	**HC**=hemisphere-capped cylinder	**HA**=hemispherical aerodisk,	**S**=flowfield structure
NV [10]=numerical study, validation with experiment of [10]	**SK**=sharp cone,	**PA**=pyramid aerodisk,	**U**=flow unsteadiness
NA=The reference was not available during the study	**SO**=sharp ogive	**PT**=pointed tip	
	TK=truncated cone	**RT**=rounded tip	
		WA=wedge aerodisk	

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